



# Product Specification

M240UAN02.3

AU OPTRONICS CORPORATION

( ) Preliminary Specification  
(V ) Final Specification

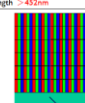
|            |                   |
|------------|-------------------|
| Module     | 24" Color TFT-LCD |
| Model Name | M240UAN02.3QY     |

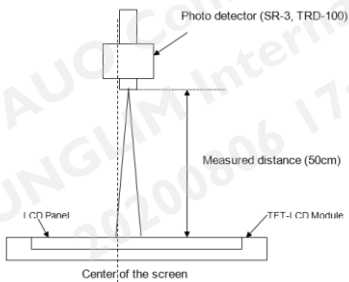
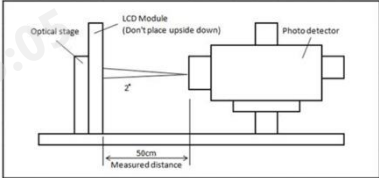
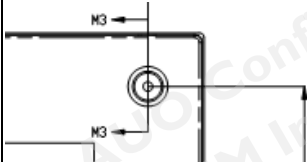
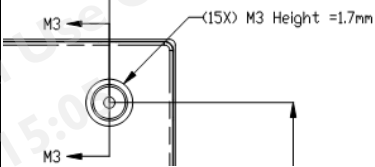
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|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|------|-------|-------|-------------|--|-------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|------|------------------------|---------------------|-------------|------|--------------------------|---------------------|
| <table><tr><td>Customer</td><td>Date</td></tr><tr><td><hr/></td><td><hr/></td></tr><tr><td>Approved by</td><td></td></tr><tr><td><hr/></td><td><hr/></td></tr></table> | Customer                 | Date | <hr/> | <hr/> | Approved by |  | <hr/> | <hr/> | <table><tr><td>Approved by</td><td>Date</td></tr><tr><td><u><b>PY Cheng</b></u></td><td><u>May 20, 2020</u></td></tr><tr><td>Prepared by</td><td>Date</td></tr><tr><td><u><b>Hannie Yeh</b></u></td><td><u>May 20, 2020</u></td></tr></table> | Approved by | Date | <u><b>PY Cheng</b></u> | <u>May 20, 2020</u> | Prepared by | Date | <u><b>Hannie Yeh</b></u> | <u>May 20, 2020</u> |
| Customer                                                                                                                                                               | Date                     |      |       |       |             |  |       |       |                                                                                                                                                                                                                                               |             |      |                        |                     |             |      |                          |                     |
| <hr/>                                                                                                                                                                  | <hr/>                    |      |       |       |             |  |       |       |                                                                                                                                                                                                                                               |             |      |                        |                     |             |      |                          |                     |
| Approved by                                                                                                                                                            |                          |      |       |       |             |  |       |       |                                                                                                                                                                                                                                               |             |      |                        |                     |             |      |                          |                     |
| <hr/>                                                                                                                                                                  | <hr/>                    |      |       |       |             |  |       |       |                                                                                                                                                                                                                                               |             |      |                        |                     |             |      |                          |                     |
| Approved by                                                                                                                                                            | Date                     |      |       |       |             |  |       |       |                                                                                                                                                                                                                                               |             |      |                        |                     |             |      |                          |                     |
| <u><b>PY Cheng</b></u>                                                                                                                                                 | <u>May 20, 2020</u>      |      |       |       |             |  |       |       |                                                                                                                                                                                                                                               |             |      |                        |                     |             |      |                          |                     |
| Prepared by                                                                                                                                                            | Date                     |      |       |       |             |  |       |       |                                                                                                                                                                                                                                               |             |      |                        |                     |             |      |                          |                     |
| <u><b>Hannie Yeh</b></u>                                                                                                                                               | <u>May 20, 2020</u>      |      |       |       |             |  |       |       |                                                                                                                                                                                                                                               |             |      |                        |                     |             |      |                          |                     |
| Note:                                                                                                                                                                  | AU Optronics corporation |      |       |       |             |  |       |       |                                                                                                                                                                                                                                               |             |      |                        |                     |             |      |                          |                     |

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## Record of Revision

| Version                                         | Date                                                  | Page                                                                                                                                      | Old description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | New Description                                                                                        | Remark                    |                                                                   |                 |      |                 |                              |                    |                       |              |       |                     |             |             |                          |                   |                |                                        |              |                |                                                          |                            |         |            |                |                                                |                |                    |       |          |               |                |                                               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| 0.1                                             | 2019/9/6                                              | All                                                                                                                                       | Preliminary Specification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                        | Preliminary Specification |                                                                   |                 |      |                 |                              |                    |                       |              |       |                     |             |             |                          |                   |                |                                        |              |                |                                                          |                            |         |            |                |                                                |                |                    |       |          |               |                |                                               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 |      |                        |                 |   |                 |                |   |                    |                           |   |                                 |  |
| 0.2                                             | 2019/10/02                                            | 32                                                                                                                                        | Thickness tolerance in drawing = +1/-0.5mm<br>(See side view)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Thickness tolerance in drawing = +/-0.5mm<br>(See side view)                                           |                           |                                                                   |                 |      |                 |                              |                    |                       |              |       |                     |             |             |                          |                   |                |                                        |              |                |                                                          |                            |         |            |                |                                                |                |                    |       |          |               |                |                                               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 |      |                        |                 |   |                 |                |   |                    |                           |   |                                 |  |
| 0.3                                             | 2019/11/22                                            | 6                                                                                                                                         | The display supports the WUXGA - 1920(H) x 1200(V) screen format and 16.7M colors (RGB 6 bits+Hi_FRC) data input).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | The display supports the WUXGA - 1920(H) x 1200(V) screen format and 16.7M colors (8 bits data input). |                           |                                                                   |                 |      |                 |                              |                    |                       |              |       |                     |             |             |                          |                   |                |                                        |              |                |                                                          |                            |         |            |                |                                                |                |                    |       |          |               |                |                                               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 |      |                        |                 |   |                 |                |   |                    |                           |   |                                 |  |
|                                                 |                                                       | 6                                                                                                                                         | <p><b>2.1 Display Characteristics</b><br/>The following items are characteristics summary on the table under 25°C condition:</p> <table><thead><tr><th>ITEMS</th><th>Unit</th><th>SPECIFICATIONS</th></tr></thead><tbody><tr><td>Screen Diagonal</td><td>[mm]</td><td>611.32 (24.07")</td></tr><tr><td>Active Area</td><td>[mm]</td><td>518.4 (H) x 324.0 (V)</td></tr><tr><td>Pixels H x V</td><td>-</td><td>1920 (H) x 1200 (V)</td></tr><tr><td>Pixel Pitch</td><td>[mm]</td><td>270 (per one triad) x270</td></tr><tr><td>Pixel Arrangement</td><td>-</td><td>R.G.B. Vertical Stripe <i>Note 2-1</i></td></tr><tr><td>Display Mode</td><td>-</td><td>AHVA Mode (Advanced Hyper-Viewing Angle), Normally Black</td></tr><tr><td>White Luminance ( Center )</td><td>[cd/m²]</td><td>250 (Typ.)</td></tr><tr><td>Contrast Ratio</td><td>-</td><td>1000 (Typ.)</td></tr><tr><td>Color Gamut</td><td>-</td><td>sRGB 95%</td></tr><tr><td>Response Time</td><td>[msec]</td><td>14 (Typ., GTG)</td></tr><tr><td>Power Consumption (LCD Module + Backlight unit)</td><td>[Watt]</td><td>14.65(Typ.)<br/>LCD module: PDD (Typ.)=4.9W @ White pattern,Fv=60Hz<br/>Backlight unit: P<sub>BLU</sub> (Typ.)=9.75W @I<sub>s</sub>=60mA</td></tr><tr><td>Weight</td><td>[Grams]</td><td>786</td></tr><tr><td>Outline Dimension</td><td>[mm]</td><td>528.2(H) x 342.4(V) x 12.5(D) Typ.</td></tr><tr><td>Electrical Interface</td><td>-</td><td>Dual channel LVDS</td></tr><tr><td>Supports Color</td><td>-</td><td>16.7M colors (RGB 6 bits+Hi_FRC)</td></tr><tr><td>Surface Treatment</td><td>-</td><td>Anti-Glare, AG25</td></tr><tr><td>Temperature Range Operating (Shipping)</td><td>[°C]</td><td>0 to +50<br/>-20 to +60</td></tr><tr><td>RoHS Compliance</td><td>-</td><td>RoHS Compliance</td></tr><tr><td>TCO Compliance</td><td>-</td><td>TCO 8.0 Compliance</td></tr><tr><td>Low Blue Light Compliance</td><td>-</td><td>LED blue chip wavelength &gt;456nm</td></tr></tbody></table> | ITEMS                                                                                                  | Unit                      | SPECIFICATIONS                                                    | Screen Diagonal | [mm] | 611.32 (24.07") | Active Area                  | [mm]               | 518.4 (H) x 324.0 (V) | Pixels H x V | -     | 1920 (H) x 1200 (V) | Pixel Pitch | [mm]        | 270 (per one triad) x270 | Pixel Arrangement | -              | R.G.B. Vertical Stripe <i>Note 2-1</i> | Display Mode | -              | AHVA Mode (Advanced Hyper-Viewing Angle), Normally Black | White Luminance ( Center ) | [cd/m²] | 250 (Typ.) | Contrast Ratio | -                                              | 1000 (Typ.)    | Color Gamut        | -     | sRGB 95% | Response Time | [msec]         | 14 (Typ., GTG)                                                    | Power Consumption (LCD Module + Backlight unit) | [Watt]                                                | 14.65(Typ.)<br>LCD module: PDD (Typ.)=4.9W @ White pattern,Fv=60Hz<br>Backlight unit: P <sub>BLU</sub> (Typ.)=9.75W @I <sub>s</sub> =60mA | Weight         | [Grams] | 786    | Outline Dimension                                       | [mm]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 528.2(H) x 342.4(V) x 12.5(D) Typ. | Electrical Interface | -     | Dual channel LVDS | Supports Color | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 16.7M colors (RGB 6 bits+Hi_FRC) | Surface Treatment  | -    | Anti-Glare, AG25 | Temperature Range Operating (Shipping) | [°C]            | 0 to +50<br>-20 to +60       | RoHS Compliance | -     | RoHS Compliance | TCO Compliance | -              | TCO 8.0 Compliance | Low Blue Light Compliance | -                  | LED blue chip wavelength >456nm | <p><b>2.1 Display Characteristics</b><br/>The following items are characteristics summary on the table under 25°C condition:</p> <table><thead><tr><th>ITEMS</th><th>Unit</th><th>SPECIFICATIONS</th></tr></thead><tbody><tr><td>Screen Diagonal</td><td>[mm]</td><td>611.32 (24.07")</td></tr><tr><td>Active Area</td><td>[mm]</td><td>518.4 (H) x 324.0 (V)</td></tr><tr><td>Pixels H x V</td><td>-</td><td>1920 (H) x 1200 (V)</td></tr><tr><td>Pixel Pitch</td><td>[mm]</td><td>270 (per one triad) x270</td></tr><tr><td>Pixel Arrangement</td><td>-</td><td>R.G.B. 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Vertical Stripe <i>Note 2-1</i> | Display Mode | -              | AHVA Mode (Advanced Hyper-Viewing Angle), Normally Black | White Luminance ( Center ) | [cd/m²] | 250 (Typ.) | Contrast Ratio | -       | 1000 (Typ.) | Response Time | [msec] | 14 (Typ., GTG) | Power Consumption (LCD Module + Backlight unit) | [Watt] | 14.65(Typ.)<br>LCD module: PDD (Typ.)=4.9W @ White pattern,Fv=60Hz<br>Backlight unit: P <sub>BLU</sub> (Typ.)=9.75W @I <sub>s</sub> =55mA | Weight | [Grams] | 2,400(Typ.) / 2,520(Max.) | Outline Dimension | [mm] | 528.2(H) x 342.4(V) x 12.5(D) Typ. | Electrical Interface | - | Dual channel LVDS | Supports Color | - | 16.7M colors (8bits RGB data input) | Surface Treatment | - | Anti-Glare, AG25 | Temperature Range Operating (Shipping) | [°C] | 0 to +50<br>-20 to +60 | RoHS Compliance | - | RoHS Compliance | TCO Compliance | - | TCO 8.0 Compliance | Low Blue Light Compliance | - | LED blue chip wavelength >452nm |  |
| ITEMS                                           | Unit                                                  | SPECIFICATIONS                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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                                  |                 |                                                       |             |       |                          |                   |                                                         |                                        |              |                |                                                          |                            |         |            |                |         |             |               |        |                |                                                 |        |                                                                                                                                           |        |         |                           |                   |      |                                    |                      |   |                   |                |   |                                     |                   |   |                  |                                        |      |                        |                 |   |                 |                |   |                    |                           |   |                                 |  |
| Screen Diagonal                                 | [mm]                                                  | 611.32 (24.07")                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| Active Area                                     | [mm]                                                  | 518.4 (H) x 324.0 (V)                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| Display Mode                                    | -                                                     | AHVA Mode (Advanced Hyper-Viewing Angle), Normally Black                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| Response Time                                   | [msec]                                                | 14 (Typ., GTG)                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| Power Consumption (LCD Module + Backlight unit) | [Watt]                                                | 14.65(Typ.)<br>LCD module: PDD (Typ.)=4.9W @ White pattern,Fv=60Hz<br>Backlight unit: P <sub>BLU</sub> (Typ.)=9.75W @I <sub>s</sub> =60mA |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| Weight                                          | [Grams]                                               | 786                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| Outline Dimension                               | [mm]                                                  | 528.2(H) x 342.4(V) x 12.5(D) Typ.                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| Electrical Interface                            | -                                                     | Dual channel LVDS                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| Screen Diagonal                                 | [mm]                                                  | 611.32 (24.07")                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| Active Area                                     | [mm]                                                  | 518.4 (H) x 324.0 (V)                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| White Luminance ( Center )                      | [cd/m²]                                               | 250 (Typ.)                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| Contrast Ratio                                  | -                                                     | 1000 (Typ.)                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| Response Time                                   | [msec]                                                | 14 (Typ., GTG)                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| Power Consumption (LCD Module + Backlight unit) | [Watt]                                                | 14.65(Typ.)<br>LCD module: PDD (Typ.)=4.9W @ White pattern,Fv=60Hz<br>Backlight unit: P <sub>BLU</sub> (Typ.)=9.75W @I <sub>s</sub> =55mA |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| Weight                                          | [Grams]                                               | 2,400(Typ.) / 2,520(Max.)                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| Outline Dimension                               | [mm]                                                  | 528.2(H) x 342.4(V) x 12.5(D) Typ.                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| Surface Treatment                               | -                                                     | Anti-Glare, AG25                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| Temperature Range Operating (Shipping)          | [°C]                                                  | 0 to +50<br>-20 to +60                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| RoHS Compliance                                 | -                                                     | RoHS Compliance                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| Low Blue Light Compliance                       | -                                                     | LED blue chip wavelength >452nm                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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L/B must be in the bottom side at normal and MNT aging state                        | Add                       |                                                                   |                 |      |                 |                              |                    |                       |              |       |                     |             |             |                          |                   |                |                                        |              |                |                                                          |                            |         |            |                |                                                |                |                    |       |          |               |                |                                                                   |                                                 |                                                       |                                                  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|                                                 |                                                       | 8                                                                                                                                         | <table><thead><tr><th>Symbol</th><th>Description</th><th>Min.</th><th>Typ.</th><th>Max.</th></tr></thead><tbody><tr><td>R<sub>x</sub></td><td rowspan="8">Color Coordinates (CIE 1931)</td><td>Red x</td><td>0.623</td><td>0.653</td><td>0.683</td></tr><tr><td>R<sub>y</sub></td><td>Red y</td><td>0.308</td><td>0.338</td><td>0.368</td></tr><tr><td>G<sub>x</sub></td><td>Green x</td><td>0.278</td><td>0.308</td><td>0.338</td></tr><tr><td>G<sub>y</sub></td><td>Green y</td><td>0.596</td><td>0.626</td><td>0.656</td></tr><tr><td>B<sub>x</sub></td><td>Blue x</td><td>0.116</td><td>0.146</td><td>0.176</td></tr><tr><td>B<sub>y</sub></td><td>Blue y</td><td>0.024</td><td>0.054</td><td>0.084</td></tr><tr><td>W<sub>x</sub></td><td>White x</td><td>0.283</td><td>0.313</td><td>0.343</td></tr><tr><td>W<sub>y</sub></td><td>White y</td><td>0.299</td><td>0.329</td><td>0.359</td></tr></tbody></table>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Symbol                                                                                                 | Description               | Min.                                                              | Typ.            | Max. | R <sub>x</sub>  | Color Coordinates (CIE 1931) | Red x              | 0.623                 | 0.653        | 0.683 | R <sub>y</sub>      | Red y       | 0.308       | 0.338                    | 0.368             | G <sub>x</sub> | Green x                                | 0.278        | 0.308          | 0.338                                                    | G <sub>y</sub>             | Green y | 0.596      | 0.626          | 0.656                                          | B <sub>x</sub> | Blue x             | 0.116 | 0.146    | 0.176         | B <sub>y</sub> | Blue y                                                            | 0.024                                           | 0.054                                                 | 0.084                                                                                                                                     | W <sub>x</sub> | White x | 0.283  | 0.313                                                   | 0.343                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | W <sub>y</sub>                     | White y              | 0.299 | 0.329             | 0.359          | <table><thead><tr><th>Symbol</th><th>Description</th><th>Min.</th><th>Typ.</th><th>Max.</th></tr></thead><tbody><tr><td>R<sub>x</sub></td><td rowspan="8">Color Coordinates (CIE 1931)</td><td>Red x</td><td>0.627</td><td>0.657</td><td>0.687</td></tr><tr><td>R<sub>y</sub></td><td>Red y</td><td>0.303</td><td>0.333</td><td>0.363</td></tr><tr><td>G<sub>x</sub></td><td>Green x</td><td>0.270</td><td>0.300</td><td>0.330</td></tr><tr><td>G<sub>y</sub></td><td>Green y</td><td>0.611</td><td>0.641</td><td>0.671</td></tr><tr><td>B<sub>x</sub></td><td>Blue x</td><td>0.114</td><td>0.144</td><td>0.174</td></tr><tr><td>B<sub>y</sub></td><td>Blue y</td><td>0.020</td><td>0.050</td><td>0.080</td></tr><tr><td>W<sub>x</sub></td><td>White x</td><td>0.283</td><td>0.313</td><td>0.343</td></tr><tr><td>W<sub>y</sub></td><td>White y</td><td>0.299</td><td>0.329</td><td>0.359</td></tr></tbody></table> | Symbol                           | Description        | Min. | Typ.             | Max.                                   | R <sub>x</sub>  | Color Coordinates (CIE 1931) | Red x           | 0.627 | 0.657           | 0.687          | R <sub>y</sub> | Red y              | 0.303                     | 0.333              | 0.363                           | G <sub>x</sub>                                                                                                                      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 | B <sub>x</sub>                                                    | Blue x          | 0.114                                                 | 0.144       | 0.174 | B <sub>y</sub>           | Blue y            | 0.020                                                   | 0.050                                  | 0.080        | W <sub>x</sub> | White x                                                  | 0.283                      | 0.313   | 0.343      | W <sub>y</sub> | White y | 0.299       | 0.329         | 0.359  |                |                                                 |        |                                                                                                                                           |        |         |                           |                   |      |                                    |                      |   |                   |                |   |                                     |                   |   |                  |    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 |      |                        |                 |   |                 |                |   |                    |                           |   |                                 |  |
| R <sub>x</sub>                                  | Color Coordinates (CIE 1931)                          | Red x                                                                                                                                     | 0.623                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            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| R <sub>y</sub>                                  |                                                       | Red y                                                                                                                                     | 0.308                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            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 |      |                        |                 |   |                 |                |   |                    |                           |   |                                 |  |
| G <sub>x</sub>                                  |                                                       | Green x                                                                                                                                   | 0.278                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            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 |      |                        |                 |   |                 |                |   |                    |                           |   |                                 |  |
| G <sub>y</sub>                                  |                                                       | Green y                                                                                                                                   | 0.596                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            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 |      |                        |                 |   |                 |                |   |                    |                           |   |                                 |  |
| B <sub>x</sub>                                  |                                                       | Blue x                                                                                                                                    | 0.116                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            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| B <sub>y</sub>                                  |                                                       | Blue y                                                                                                                                    | 0.024                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            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 |      |                        |                 |   |                 |                |   |                    |                           |   |                                 |  |
| W <sub>x</sub>                                  |                                                       | White x                                                                                                                                   | 0.283                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            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| W <sub>y</sub>                                  |                                                       | White y                                                                                                                                   | 0.299                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            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| Symbol                                          | Description                                           | Min.                                                                                                                                      | Typ.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             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| R <sub>x</sub>                                  | Color Coordinates (CIE 1931)                          | Red x                                                                                                                                     | 0.627                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            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| R <sub>y</sub>                                  |                                                       | Red y                                                                                                                                     | 0.303                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            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 |      |                        |                 |   |                 |                |   |                    |                           |   |                                 |  |
| G <sub>x</sub>                                  |                                                       | Green x                                                                                                                                   | 0.270                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            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| G <sub>y</sub>                                  |                                                       | Green y                                                                                                                                   | 0.611                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            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| B <sub>x</sub>                                  |                                                       | Blue x                                                                                                                                    | 0.114                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            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| B <sub>y</sub>                                  |                                                       | Blue y                                                                                                                                    | 0.020                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            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| W <sub>x</sub>                                  |                                                       | White x                                                                                                                                   | 0.283                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            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| W <sub>y</sub>                                  |                                                       | White y                                                                                                                                   | 0.299                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            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|                                                 |                                                       | 28                                                                                                                                        | <p><b>4.3.1 Absolute Maximum Rating</b><br/>Permanent damage may occur if exceeding the following maximum rating.</p> <table><thead><tr><th>Symbol</th><th>Description</th><th>Min</th><th>Max</th><th>Unit</th><th>Remark</th></tr></thead><tbody><tr><td>I<sub>s</sub></td><td>LED String Current</td><td>0</td><td>150</td><td>[mA]</td><td>100% duty ratio</td></tr></tbody></table> <p><b>4.3.2 Recommended Operating Condition</b></p> <table><thead><tr><th>Symbol</th><th>Description</th><th>Min.</th><th>Typ.</th><th>Max.</th><th>Unit</th><th>Remark</th></tr></thead><tbody><tr><td>I<sub>s</sub></td><td>LED String Current</td><td>-</td><td>55</td><td>60</td><td>[mA]</td><td>100% duty ratio of LED chip<br/><i>Note 4-7</i></td></tr><tr><td>V<sub>s</sub></td><td>LED String Voltage</td><td>42</td><td>44.3</td><td>46.5</td><td>[Volt]</td><td>I<sub>s</sub>=60mA @ 100% duty ratio; <i>Note 4-1, Note 4-5</i></td></tr><tr><td>ΔV<sub>s</sub></td><td>Maximum V<sub>s</sub> Voltage Deviation of light bar</td><td>-</td><td>-</td><td>1.5</td><td>[Volt]</td><td>I<sub>s</sub>=60mA @ 100% duty ratio; <i>Note 4-2</i></td></tr></tbody></table>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Symbol                                                                                                 | Description               | Min                                                               | Max             | Unit | Remark          | I <sub>s</sub>               | LED String Current | 0                     | 150          | [mA]  | 100% duty ratio     | Symbol      | Description | Min.                     | Typ.              | Max.           | Unit                                   | Remark       | I <sub>s</sub> | LED String Current                                       | -                          | 55      | 60         | [mA]           | 100% duty ratio of LED chip<br><i>Note 4-7</i> | V <sub>s</sub> | LED String Voltage | 42    | 44.3     | 46.5          | [Volt]         | I <sub>s</sub> =60mA @ 100% duty ratio; <i>Note 4-1, Note 4-5</i> | ΔV <sub>s</sub>                                 | Maximum V <sub>s</sub> Voltage Deviation of light bar | -                                                                                                                                         | -              | 1.5     | [Volt] | I <sub>s</sub> =60mA @ 100% duty ratio; <i>Note 4-2</i> | <p><b>4.3.1 Absolute Maximum Rating</b><br/>Permanent damage may occur if exceeding the following maximum rating.</p> <table><thead><tr><th>Symbol</th><th>Description</th><th>Min</th><th>Max</th><th>Unit</th><th>Remark</th></tr></thead><tbody><tr><td>I<sub>s</sub></td><td>LED String Current</td><td>0</td><td>120</td><td>[mA]</td><td>100% duty ratio</td></tr></tbody></table> <p><b>4.3.2 Recommended Operating Condition</b></p> <table><thead><tr><th>Symbol</th><th>Description</th><th>Min.</th><th>Typ.</th><th>Max.</th><th>Unit</th><th>Remark</th></tr></thead><tbody><tr><td>I<sub>s</sub></td><td>LED String Current</td><td>-</td><td>55</td><td>60</td><td>[mA]</td><td>100% duty ratio of LED chip<br/><i>Note 4-7</i></td></tr><tr><td>V<sub>s</sub></td><td>LED String Voltage</td><td>42</td><td>44.3</td><td>46.5</td><td>[Volt]</td><td>I<sub>s</sub>=55mA @ 100% duty ratio; <i>Note 4-1, Note 4-5</i></td></tr><tr><td>ΔV<sub>s</sub></td><td>Maximum V<sub>s</sub> Voltage Deviation of light bar</td><td>-</td><td>-</td><td>3</td><td>[Volt]</td><td>I<sub>s</sub>=55mA @ 100% duty ratio; <i>Note 4-2</i></td></tr></tbody></table> | Symbol                             | Description          | Min   | Max               | Unit           | Remark                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | I <sub>s</sub>                   | LED String Current | 0    | 120              | [mA]                                   | 100% duty ratio | Symbol                       | Description     | Min.  | Typ.            | Max.           | Unit           | Remark             | I <sub>s</sub>            | LED String Current | -                               | 55                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 60      | [mA]  | 100% duty ratio of LED chip<br><i>Note 4-7</i> | V <sub>s</sub>  | LED String Voltage | 42              | 44.3        | 46.5  | [Volt]                | I <sub>s</sub> =55mA @ 100% duty ratio; <i>Note 4-1, Note 4-5</i> | ΔV <sub>s</sub> | Maximum V <sub>s</sub> Voltage Deviation of light bar | -           | -     | 3                        | [Volt]            | I <sub>s</sub> =55mA @ 100% duty ratio; <i>Note 4-2</i> |                                        |              |                |                                                          |                            |         |            |                |         |             |               |        |                |                                                 |        |                                                                                                                                           |        |         |                           |                   |      |                                    |                      |   |                   |                |   |                                     |                   |   |                  |                                        |      |                        |                 |   |                 |                |   |                    |                           |   |                                 |  |
| Symbol                                          | Description                                           | Min                                                                                                                                       | Max                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| V <sub>s</sub>                                  | LED String Voltage                                    | 42                                                                                                                                        | 44.3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             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| ΔV <sub>s</sub>                                 | Maximum V <sub>s</sub> Voltage Deviation of light bar | -                                                                                                                                         | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                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| I <sub>s</sub>                                  | LED String Current                                    | -                                                                                                                                         | 55                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 60                                                                                                     | [mA]                      | 100% duty ratio of LED chip<br><i>Note 4-7</i>                    |                 |      |                 |                              |                    |                       |              |       |                     |             |             |                          |                   |                |                                        |              |                |                                                          |                            |         |            |                |                                                |                |                    |       |          |               |                |                                               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| V <sub>s</sub>                                  | LED String Voltage                                    | 42                                                                                                                                        | 44.3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 46.5                                                                                                   | [Volt]                    | I <sub>s</sub> =55mA @ 100% duty ratio; <i>Note 4-1, Note 4-5</i> |                 |      |                 |                              |                    |                       |              |       |                     |             |             |                          |                   |                |                                        |              |                |                                                          |                            |         |            |                |                                                |                |                    |       |          |               |                |                                               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| ΔV <sub>s</sub>                                 | Maximum V <sub>s</sub> Voltage Deviation of light bar | -                                                                                                                                         | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                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|                                                 |                                                       | 30                                                                                                                                        | Note 4-4:<br>Definition of life time:<br>b. Test condition: I <sub>s</sub> = 65mA and                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Note 4-4:<br>Definition of life time:<br>b. Test condition: I <sub>s</sub> = 55mA and                  |                           |                                                                   |                 |      |                 |                              |                    |                       |              |       |                     |             |             |                          |                   |                |                                        |              |                |                                                          |                            |         |            |                |                                                |                |                    |       |          |               |                |                                                                   |                                                 |                                                       |                                                                           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|                      |                                                     | 32                                 | Update Backview Drawing                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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|                      |                                                     | 35                                 | <div>8.2 Pallet and shipment information</div> <table><tr><th>Item</th><th>Q'ty</th><th>Specification</th><th>Weight(kg)</th><th>Remark</th></tr><tr><td>Panel</td><td>1</td><td>528.2(H) × 342.4(V) × 12.5(D) Typ.</td><td>TBD</td><td></td></tr><tr><td>Cushion</td><td>1</td><td>-</td><td>0.6</td><td></td></tr><tr><td>Box</td><td>1</td><td>619(L)mm x 373(W)mm x 427(H)mm</td><td>1.5</td><td>without Panel &amp; cushion</td></tr><tr><td>Packing Box</td><td>11 pcs/Box</td><td>619(L)mm x 373(W)mm x 427(H)mm</td><td>29.93</td><td>with panel &amp; cushion</td></tr><tr><td>Pallet</td><td>1</td><td>1315(L)mm x 1150(W)mm x 132(H)mm</td><td>17.00</td><td></td></tr><tr><td>Pallet after Packing</td><td>18 boxes/pallet</td><td>1315(L)mm x 1150(W)mm x 132(H)mm</td><td>555.74</td><td></td></tr></table>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Item                                                                                                                                                 | Q'ty                 | Specification                                | Weight(kg) | Remark | Panel | 1      | 528.2(H) × 342.4(V) × 12.5(D) Typ. | TBD                                                 |      | Cushion | 1    | -                    | 0.6                                     |                | Box                                                 | 1    | 619(L)mm x 373(W)mm x 427(H)mm | 1.5  | without Panel & cushion | Packing Box                                  | 11 pcs/Box | 619(L)mm x 373(W)mm x 427(H)mm                    | 29.93 | with panel & cushion | Pallet | 1   | 1315(L)mm x 1150(W)mm x 132(H)mm   | 17.00            |                                                   | Pallet after Packing | 18 boxes/pallet | 1315(L)mm x 1150(W)mm x 132(H)mm | 555.74   |                     | <div>8.2 Pallet and shipment information</div> <table><tr><th>Item</th><th>Q'ty</th><th>Specification</th><th>Weight(kg)</th><th>Remark</th></tr><tr><td>Panel</td><td>1</td><td>528.2(H) × 342.4(V) × 12.5(D) Typ.</td><td>2.4</td><td></td></tr><tr><td>Cushion</td><td>1</td><td>-</td><td>0.6</td><td></td></tr><tr><td>Box</td><td>1</td><td>619(L)mm x 373(W)mm x 427(H)mm</td><td>1.5</td><td>without Panel &amp; cushion</td></tr><tr><td>Packing Box</td><td>11 pcs/Box</td><td>619(L)mm x 373(W)mm x 427(H)mm</td><td>28.5</td><td>with panel &amp; cushion</td></tr><tr><td>Pallet</td><td>1</td><td>1315(L)mm x 1150(W)mm x 132(H)mm</td><td>17.00</td><td></td></tr><tr><td>Pallet after Packing</td><td>18 boxes/pallet</td><td>1315(L)mm x 1150(W)mm x 132(H)mm</td><td>530</td><td></td></tr></table> | Item                              | Q'ty       | Specification | Weight(kg) | Remark   | Panel          | 1                                  | 528.2(H) × 342.4(V) × 12.5(D) Typ.               | 2.4                      |     | Cushion  | 1              | -                                | 0.6                                                                                                                                                                                                                                                                                                                                                                                                                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                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 11 pcs/Box | 619(L)mm x 373(W)mm x 427(H)mm | 28.5           | with panel & cushion              | Pallet     | 1        | 1315(L)mm x 1150(W)mm x 132(H)mm        | 17.00       |                                                            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| Panel                | 1                                                   | 528.2(H) × 342.4(V) × 12.5(D) Typ. | TBD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| Cushion              | 1                                                   | -                                  | 0.6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| Box                  | 1                                                   | 619(L)mm x 373(W)mm x 427(H)mm     | 1.5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| Packing Box          | 11 pcs/Box                                          | 619(L)mm x 373(W)mm x 427(H)mm     | 29.93                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                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| Pallet               | 1                                                   | 1315(L)mm x 1150(W)mm x 132(H)mm   | 17.00                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                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| Pallet after Packing | 18 boxes/pallet                                     | 1315(L)mm x 1150(W)mm x 132(H)mm   | 555.74                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               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| Item                 | Q'ty                                                | Specification                      | Weight(kg)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           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| Panel                | 1                                                   | 528.2(H) × 342.4(V) × 12.5(D) Typ. | 2.4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| Cushion              | 1                                                   | -                                  | 0.6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| Box                  | 1                                                   | 619(L)mm x 373(W)mm x 427(H)mm     | 1.5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| Packing Box          | 11 pcs/Box                                          | 619(L)mm x 373(W)mm x 427(H)mm     | 28.5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 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| Pallet               | 1                                                   | 1315(L)mm x 1150(W)mm x 132(H)mm   | 17.00                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                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| Pallet after Packing | 18 boxes/pallet                                     | 1315(L)mm x 1150(W)mm x 132(H)mm   | 530                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| 0.5                  | 2020/3/10                                           | 8                                  | <table><tr><th>Symbol</th><th>Description</th><th>Min.</th><th>Typ.</th><th>Max.</th><th>Unit</th><th>Remark</th></tr><tr><td>L<sub>w</sub></td><td>White Luminance (Center of screen)</td><td>200</td><td>250</td><td>-</td><td>[cd/m<sup>2</sup>]</td><td>Note 2-2<br/>By SR-3</td></tr><tr><td>L<sub>u</sub></td><td>Luminance Uniformity (9 points)</td><td>75</td><td>80</td><td>-</td><td>[%]</td><td>Note 2-3<br/>By SR-3</td></tr><tr><td>CR</td><td>Contrast Ratio (Center of screen)</td><td>600</td><td>1000</td><td>-</td><td>-</td><td>Note 2-4<br/>By SR-3</td></tr><tr><td>θ<sub>h</sub></td><td>Horizontal Viewing Angle<br/>(CR&gt;10)</td><td>Right:<br/>75°</td><td>89°</td><td>-</td><td>[degree]</td><td rowspan="6">Note 2-4<br/>By SR-3</td></tr><tr><td>θ<sub>v</sub></td><td>Vertical Viewing Angle<br/>(CR&gt;10)</td><td>Up:<br/>70°</td><td>89°</td><td>-</td><td>[degree]</td></tr><tr><td>θ<sub>h</sub></td><td>Horizontal Viewing Angle<br/>(CR&gt;5)</td><td>Right:<br/>75°</td><td>89°</td><td>-</td><td>[degree]</td></tr><tr><td>θ<sub>v</sub></td><td>Vertical Viewing Angle<br/>(CR&gt;5)</td><td>Up:<br/>70°</td><td>89°</td><td>-</td><td>[degree]</td></tr><tr><td>θ<sub>h</sub></td><td>Horizontal Viewing Angle<br/>(CR&gt;10)</td><td>Right:<br/>75°</td><td>89°</td><td>-</td><td>[degree]</td></tr><tr><td>θ<sub>v</sub></td><td>Vertical Viewing Angle<br/>(CR&gt;10)</td><td>Up:<br/>70°</td><td>89°</td><td>-</td><td>[degree]</td></tr></table> | Symbol                                                                                                                                               | Description          | Min.                                         | Typ.       | Max.   | Unit  | Remark | L <sub>w</sub>                     | White Luminance (Center of screen)                  | 200  | 250     | -    | [cd/m <sup>2</sup> ] | Note 2-2<br>By SR-3                     | L <sub>u</sub> | Luminance Uniformity (9 points)                     | 75   | 80                             | -    | [%]                     | Note 2-3<br>By SR-3                          | CR         | Contrast Ratio (Center of screen)                 | 600   | 1000                 | -      | -   | Note 2-4<br>By SR-3                | θ <sub>h</sub>   | Horizontal Viewing Angle<br>(CR>10)               | Right:<br>75°        | 89°             | -                                | [degree] | Note 2-4<br>By SR-3 | θ <sub>v</sub>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Vertical Viewing Angle<br>(CR>10) | Up:<br>70° | 89°           | -          | [degree] | θ <sub>h</sub> | Horizontal Viewing Angle<br>(CR>5) | Right:<br>75°                                    | 89°                      | -   | [degree] | θ <sub>v</sub> | Vertical Viewing Angle<br>(CR>5) | Up:<br>70°                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 89°                                              | -           | [degree] | θ <sub>h</sub>                 | Horizontal Viewing Angle<br>(CR>10) | Right:<br>75°           | 89°                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | -          | [degree]                       | θ <sub>v</sub> | Vertical Viewing Angle<br>(CR>10) | Up:<br>70° | 89°      | -                                       | [degree]    | <table><tr><th>Symbol</th><th>Description</th><th>Min.</th><th>Typ.</th><th>Max.</th><th>Unit</th><th>Remark</th></tr><tr><td>L<sub>w</sub></td><td>White Luminance (Center of screen)</td><td>200</td><td>250</td><td>-</td><td>[cd/m<sup>2</sup>]</td><td>Note 2-2<br/>By SR-3</td></tr><tr><td>L<sub>u</sub></td><td>Luminance Uniformity (9 points)</td><td>75</td><td>80</td><td>-</td><td>[%]</td><td>Note 2-3<br/>By SR-3</td></tr><tr><td>CR</td><td>Contrast Ratio (Center of screen)</td><td>700</td><td>1000</td><td>-</td><td>-</td><td>Note 2-4<br/>By SR-3</td></tr><tr><td>θ<sub>h</sub></td><td>Horizontal Viewing Angle<br/>(CR&gt;10)</td><td>Right:<br/>85°</td><td>89°</td><td>-</td><td>[degree]</td><td rowspan="6">Note 2-4<br/>By SR-3</td></tr><tr><td>θ<sub>v</sub></td><td>Vertical Viewing Angle<br/>(CR&gt;10)</td><td>Up:<br/>85°</td><td>89°</td><td>-</td><td>[degree]</td></tr><tr><td>θ<sub>h</sub></td><td>Horizontal Viewing Angle<br/>(CR&gt;5)</td><td>Right:<br/>85°</td><td>89°</td><td>-</td><td>[degree]</td></tr><tr><td>θ<sub>v</sub></td><td>Vertical Viewing Angle<br/>(CR&gt;5)</td><td>Up:<br/>85°</td><td>89°</td><td>-</td><td>[degree]</td></tr><tr><td>θ<sub>h</sub></td><td>Horizontal Viewing Angle<br/>(CR&gt;10)</td><td>Right:<br/>85°</td><td>89°</td><td>-</td><td>[degree]</td></tr><tr><td>θ<sub>v</sub></td><td>Vertical Viewing Angle<br/>(CR&gt;10)</td><td>Up:<br/>85°</td><td>89°</td><td>-</td><td>[degree]</td></tr></table> | Symbol               | Description     | Min.                             | Typ.     | Max.                                         | Unit      | Remark                                            | L <sub>w</sub> | White Luminance (Center of screen) | 200 | 250      | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | [cd/m <sup>2</sup> ] | Note 2-2<br>By SR-3                               | L <sub>u</sub> | Luminance Uniformity (9 points) | 75    | 80     | -        | [%]               | Note 2-3<br>By SR-3                              | CR     | Contrast Ratio (Center of screen) | 700  | 1000   | -        | -         | Note 2-4<br>By SR-3                              | θ <sub>h</sub>           | Horizontal Viewing Angle<br>(CR>10) | Right:<br>85° | 89° | -        | [degree] | Note 2-4<br>By SR-3                              | θ <sub>v</sub> | Vertical Viewing Angle<br>(CR>10) | Up:<br>85° | 89° | - | [degree] | θ <sub>h</sub> | Horizontal Viewing Angle<br>(CR>5) | Right:<br>85° | 89° | -  | [degree] | θ <sub>v</sub> | Vertical Viewing Angle<br>(CR>5) | Up:<br>85° | 89°  | -    | [degree] | θ <sub>h</sub> | Horizontal Viewing Angle<br>(CR>10) | Right:<br>85° | 89° | -   | [degree] | θ <sub>v</sub> | Vertical Viewing Angle<br>(CR>10) | Up:<br>85° | 89°      | -  | [degree] |     |      |   |    |           |      |      |      |     |          |      |        |      |      |      |    |          |      |           |      |      |      |     |          |  |
| Symbol               | Description                                         | Min.                               | Typ.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 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| L <sub>w</sub>       | White Luminance (Center of screen)                  | 200                                | 250                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| L <sub>u</sub>       | Luminance Uniformity (9 points)                     | 75                                 | 80                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   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| CR                   | Contrast Ratio (Center of screen)                   | 600                                | 1000                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 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| θ <sub>h</sub>       | Horizontal Viewing Angle<br>(CR>10)                 | Right:<br>75°                      | 89°                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| θ <sub>v</sub>       | Vertical Viewing Angle<br>(CR>10)                   | Up:<br>70°                         | 89°                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| θ <sub>h</sub>       | Horizontal Viewing Angle<br>(CR>5)                  | Right:<br>75°                      | 89°                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| θ <sub>v</sub>       | Vertical Viewing Angle<br>(CR>5)                    | Up:<br>70°                         | 89°                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| θ <sub>h</sub>       | Horizontal Viewing Angle<br>(CR>10)                 | Right:<br>75°                      | 89°                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| θ <sub>v</sub>       | Vertical Viewing Angle<br>(CR>10)                   | Up:<br>70°                         | 89°                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| Symbol               | Description                                         | Min.                               | Typ.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 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| L <sub>w</sub>       | White Luminance (Center of screen)                  | 200                                | 250                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| L <sub>u</sub>       | Luminance Uniformity (9 points)                     | 75                                 | 80                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   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| CR                   | Contrast Ratio (Center of screen)                   | 700                                | 1000                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 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| θ <sub>h</sub>       | Horizontal Viewing Angle<br>(CR>10)                 | Right:<br>85°                      | 89°                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| θ <sub>v</sub>       | Vertical Viewing Angle<br>(CR>10)                   | Up:<br>85°                         | 89°                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| θ <sub>h</sub>       | Horizontal Viewing Angle<br>(CR>5)                  | Right:<br>85°                      | 89°                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| θ <sub>v</sub>       | Vertical Viewing Angle<br>(CR>5)                    | Up:<br>85°                         | 89°                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| θ <sub>h</sub>       | Horizontal Viewing Angle<br>(CR>10)                 | Right:<br>85°                      | 89°                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| θ <sub>v</sub>       | Vertical Viewing Angle<br>(CR>10)                   | Up:<br>85°                         | 89°                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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|                      |                                                     | 9                                  | <div>Note 2-2: Equipment setup:</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                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|                      |                                                     | 14                                 | <table><tr><th>Symbol</th><th>Description</th><th>Min.</th><th>Typ.</th><th>Max.</th><th>Unit</th><th>Remark</th></tr><tr><td>20- RxCCLK+</td><td>Negative LVDS differential clock input (Even clock)</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td></tr><tr><td>21- RxCCLK+</td><td>Positive LVDS differential clock input (Even clock)</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td></tr><tr><td>22- RxE3+</td><td>Negative LVDS differential data input (Even data)</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td></tr><tr><td>23- RxE3+</td><td>Positive LVDS differential data input (Even data)</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td></tr><tr><td>24- GND</td><td>Ground</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td></tr><tr><td>25- NC</td><td>No connection for AUO test only. Do not connect.</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td></tr><tr><td>26- NC</td><td>No connection for AUO test only. Do not connect.</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td></tr></table>                                                                                                                                                                                                                                                                                                                                                                                                                               | Symbol                                                                                                                                               | Description          | Min.                                         | Typ.       | Max.   | Unit  | Remark | 20- RxCCLK+                        | Negative LVDS differential clock input (Even clock) | -    | -       | -    | -                    | -                                       | 21- RxCCLK+    | Positive LVDS differential clock input (Even clock) | -    | -                              | -    | -                       | -                                            | 22- RxE3+  | Negative LVDS differential data input (Even data) | -     | -                    | -      | -   | -                                  | 23- RxE3+        | Positive LVDS differential data input (Even data) | -                    | -               | -                                | -        | -                   | 24- GND                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Ground                            | -          | -             | -          | -        | -              | 25- NC                             | No connection for AUO test only. Do not connect. | -                        | -   | -        | -              | -                                | 26- NC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | No connection for AUO test only. Do not connect. | -           | -        | -                              | -                                   | -                       | <table><tr><th>Symbol</th><th>Description</th><th>Min.</th><th>Typ.</th><th>Max.</th><th>Unit</th><th>Remark</th></tr><tr><td>21- RxCCLK+</td><td>Positive LVDS differential clock input (Even clock)</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td></tr><tr><td>22- RxE3+</td><td>Negative LVDS differential data input (Even data)</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td></tr><tr><td>23- RxE3+</td><td>Positive LVDS differential data input (Even data)</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td></tr><tr><td>24- NC</td><td>No connection for AUO test only. Do not connect.</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td></tr><tr><td>25- NC</td><td>No connection for AUO test only. Do not connect.</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td></tr><tr><td>26- NC</td><td>No connection for AUO test only. Do not connect.</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td></tr></table> | Symbol     | Description                    | Min.           | Typ.                              | Max.       | Unit     | Remark                                  | 21- RxCCLK+ | Positive LVDS differential clock input (Even clock)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | -                    | -               | -                                | -        | -                                            | 22- RxE3+ | Negative LVDS differential data input (Even data) | -              | -                                  | -   | -        | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 23- RxE3+            | Positive LVDS differential data input (Even data) | -              | -                               | -     | -      | -        | 24- NC            | No connection for AUO test only. Do not connect. | -      | -                                 | -    | -      | -        | 25- NC    | No connection for AUO test only. Do not connect. | -                        | -                                   | -             | -   | -        | 26- NC   | No connection for AUO test only. Do not connect. | -              | -                                 | -          | -   | - |          |                |                                    |               |     |    |          |                |                                  |            |      |      |          |                |                                     |               |     |     |          |                |                                   |            |          |    |          |     |      |   |    |           |      |      |      |     |          |      |        |      |      |      |    |          |      |           |      |      |      |     |          |  |
| Symbol               | Description                                         | Min.                               | Typ.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 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| 20- RxCCLK+          | Negative LVDS differential clock input (Even clock) | -                                  | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    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| 21- RxCCLK+          | Positive LVDS differential clock input (Even clock) | -                                  | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    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| 22- RxE3+            | Negative LVDS differential data input (Even data)   | -                                  | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    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| 23- RxE3+            | Positive LVDS differential data input (Even data)   | -                                  | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    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| 24- GND              | Ground                                              | -                                  | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    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| 25- NC               | No connection for AUO test only. Do not connect.    | -                                  | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 26- NC               | No connection for AUO test only. Do not connect.    | -                                  | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| Symbol               | Description                                         | Min.                               | Typ.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 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| 21- RxCCLK+          | Positive LVDS differential clock input (Even clock) | -                                  | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    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| 22- RxE3+            | Negative LVDS differential data input (Even data)   | -                                  | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    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| 23- RxE3+            | Positive LVDS differential data input (Even data)   | -                                  | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    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| 24- NC               | No connection for AUO test only. Do not connect.    | -                                  | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 25- NC               | No connection for AUO test only. Do not connect.    | -                                  | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 26- NC               | No connection for AUO test only. Do not connect.    | -                                  | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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|                      |                                                     | 21                                 | <table><tr><th>Symbol</th><th>Description</th><th>Min.</th><th>Typ.</th><th>Max.</th><th>Unit</th><th>Remark</th></tr><tr><td>Tp</td><td>Period</td><td>1214</td><td>1235</td><td>1965</td><td>Th</td><td>-</td></tr><tr><td>Tdisp (v)</td><td>Active</td><td>1200</td><td>1200</td><td>1200</td><td>Th</td><td>-</td></tr><tr><td>Tblk (v)</td><td>Blanking</td><td>14</td><td>35</td><td>765</td><td>Th</td><td>-</td></tr><tr><td>Fv</td><td>Frequency</td><td>49</td><td>60</td><td>75</td><td>Hz</td><td>-</td></tr><tr><td>Th</td><td>Period</td><td>1000</td><td>1040</td><td>1428</td><td>Tclk</td><td>-</td></tr><tr><td>Tdisp (h)</td><td>Active</td><td>960</td><td>960</td><td>960</td><td>Tclk</td><td>-</td></tr><tr><td>Tblk (h)</td><td>Blanking</td><td>40</td><td>80</td><td>468</td><td>Tclk</td><td>-</td></tr><tr><td>Fh</td><td>Frequency</td><td>59.5</td><td>74.0</td><td>96.3</td><td>KHz</td><td>Note 3-3</td></tr><tr><td>Tclk</td><td>Period</td><td>10.4</td><td>13.0</td><td>16.8</td><td>ns</td><td>Note 3-4</td></tr><tr><td>Fclk</td><td>Frequency</td><td>59.5</td><td>77.0</td><td>97.5</td><td>MHz</td><td>Note 3-4</td></tr></table>                                                                                                                                                                                                                                                                                                                        | Symbol                                                                                                                                               | Description          | Min.                                         | Typ.       | Max.   | Unit  | Remark | Tp                                 | Period                                              | 1214 | 1235    | 1965 | Th                   | -                                       | Tdisp (v)      | Active                                              | 1200 | 1200                           | 1200 | Th                      | -                                            | Tblk (v)   | Blanking                                          | 14    | 35                   | 765    | Th  | -                                  | Fv               | Frequency                                         | 49                   | 60              | 75                               | Hz       | -                   | Th                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Period                            | 1000       | 1040          | 1428       | Tclk     | -              | Tdisp (h)                          | Active                                           | 960                      | 960 | 960      | Tclk           | -                                | Tblk (h)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Blanking                                         | 40          | 80       | 468                            | Tclk                                | -                       | Fh                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Frequency  | 59.5                           | 74.0           | 96.3                              | KHz        | Note 3-3 | Tclk                                    | Period      | 10.4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 13.0                 | 16.8            | ns                               | Note 3-4 | Fclk                                         | Frequency | 59.5                                              | 77.0           | 97.5                               | MHz | Note 3-4 | <table><tr><th>Symbol</th><th>Description</th><th>Min.</th><th>Typ.</th><th>Max.</th><th>Unit</th><th>Remark</th></tr><tr><td>Tp</td><td>Period</td><td>1214</td><td>1235</td><td>1965</td><td>Th</td><td>-</td></tr><tr><td>Tdisp (v)</td><td>Active</td><td>1200</td><td>1200</td><td>1200</td><td>Th</td><td>-</td></tr><tr><td>Tblk (v)</td><td>Blanking</td><td>14</td><td>35</td><td>765</td><td>Th</td><td>-</td></tr><tr><td>Fv</td><td>Frequency</td><td>49</td><td>60</td><td>75</td><td>Hz</td><td>-</td></tr><tr><td>Th</td><td>Period</td><td>1000</td><td>1040</td><td>1428</td><td>Tclk</td><td>-</td></tr><tr><td>Tdisp (h)</td><td>Active</td><td>960</td><td>960</td><td>960</td><td>Tclk</td><td>-</td></tr><tr><td>Tblk (h)</td><td>Blanking</td><td>40</td><td>80</td><td>468</td><td>Tclk</td><td>-</td></tr><tr><td>Fh</td><td>Frequency</td><td>59.5</td><td>74.0</td><td>96.3</td><td>KHz</td><td>Note 3-3</td></tr><tr><td>Tclk</td><td>Period</td><td>10.4</td><td>13.0</td><td>16.8</td><td>ns</td><td>Note 3-4</td></tr><tr><td>Fclk</td><td>Frequency</td><td>59.5</td><td>77.0</td><td>97.5</td><td>MHz</td><td>Note 3-4</td></tr></table> | Symbol               | Description                                       | Min.           | Typ.                            | Max.  | Unit   | Remark   | Tp                | Period                                           | 1214   | 1235                              | 1965 | Th     | -        | Tdisp (v) | Active                                           | 1200                     | 1200                                | 1200          | Th  | -        | Tblk (v) | Blanking                                         | 14             | 35                                | 765        | Th  | - | Fv       | Frequency      | 49                                 | 60            | 75  | Hz | -        | Th             | Period                           | 1000       | 1040 | 1428 | Tclk     | -              | Tdisp (h)                           | Active        | 960 | 960 | 960      | Tclk           | -                                 | Tblk (h)   | Blanking | 40 | 80       | 468 | Tclk | - | Fh | Frequency | 59.5 | 74.0 | 96.3 | KHz | Note 3-3 | Tclk | Period | 10.4 | 13.0 | 16.8 | ns | Note 3-4 | Fclk | Frequency | 59.5 | 77.0 | 97.5 | MHz | Note 3-4 |  |
| Symbol               | Description                                         | Min.                               | Typ.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 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| Tdisp (v)            | Active                                              | 1200                               | 1200                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 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| Tblk (v)             | Blanking                                            | 14                                 | 35                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   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| Fv                   | Frequency                                           | 49                                 | 60                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   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| Fh                   | Frequency                                           | 59.5                               | 74.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 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| Tclk                 | Period                                              | 10.4                               | 13.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 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| Fclk                 | Frequency                                           | 59.5                               | 77.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 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| Tp                   | Period                                              | 1214                               | 1235                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 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| Tdisp (v)            | Active                                              | 1200                               | 1200                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 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| Tblk (v)             | Blanking                                            | 14                                 | 35                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   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| Fv                   | Frequency                                           | 49                                 | 60                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   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| Tdisp (h)            | Active                                              | 960                                | 960                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| Tblk (h)             | Blanking                                            | 40                                 | 80                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   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| Fh                   | Frequency                                           | 59.5                               | 74.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 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| Tclk                 | Period                                              | 10.4                               | 13.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 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| Fclk                 | Frequency                                           | 59.5                               | 77.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 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| 1.0                  |                                                     | 29                                 | <div>4.3.2 Recommended Operating Condition</div> <div>(Ta=25°C)</div> <table><tr><th>Symbol</th><th>Description</th><th>Min.</th><th>Typ.</th><th>Max.</th><th>Unit</th><th>Remark</th></tr><tr><td>Is</td><td>LED String Current</td><td>-</td><td>55</td><td>60</td><td>[mA]</td><td>100% duty ratio of LED chip<br/>Note 4-2</td></tr><tr><td>Vs</td><td>LED String Voltage</td><td>42</td><td>44.3</td><td>46.5</td><td>[V]</td><td>I=55mA @ 100% duty ratio; Note 4-1, Note 4-3</td></tr><tr><td>ΔVs</td><td>Maximum Vs Voltage Deviation of light bar</td><td>-</td><td>-</td><td>3</td><td>[V]</td><td>I=55mA @ 100% duty ratio; Note 4-2</td></tr><tr><td>P<sub>LED</sub></td><td>LED Light Bar Power Consumption</td><td>-</td><td>9.75</td><td>10.23</td><td>[Watt]</td><td>Note 4-3</td></tr><tr><td>LT<sub>LED</sub></td><td>LED Life Time</td><td>30,000</td><td>-</td><td>-</td><td>[Hour]</td><td>Note 4-4</td></tr><tr><td>OVP</td><td>Over Voltage Protection in system board</td><td>110%<br/>V<sub>max</sub></td><td>-</td><td>-</td><td>[V]</td><td>Note 4-5</td></tr></table>                                                                                                                                                                                                                                                                                                                                                                                               | Symbol                                                                                                                                               | Description          | Min.                                         | Typ.       | Max.   | Unit  | Remark | Is                                 | LED String Current                                  | -    | 55      | 60   | [mA]                 | 100% duty ratio of LED chip<br>Note 4-2 | Vs             | LED String Voltage                                  | 42   | 44.3                           | 46.5 | [V]                     | I=55mA @ 100% duty ratio; Note 4-1, Note 4-3 | ΔVs        | Maximum Vs Voltage Deviation of light bar         | -     | -                    | 3      | [V] | I=55mA @ 100% duty ratio; Note 4-2 | P <sub>LED</sub> | LED Light Bar Power Consumption                   | -                    | 9.75            | 10.23                            | [Watt]   | Note 4-3            | LT <sub>LED</sub>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | LED Life Time                     | 30,000     | -             | -          | [Hour]   | Note 4-4       | OVP                                | Over Voltage Protection in system board          | 110%<br>V <sub>max</sub> | -   | -        | [V]            | Note 4-5                         | <div>4.3.2 Recommended Operating Condition</div> <div>(Ta=25°C)</div> <table><tr><th>Symbol</th><th>Description</th><th>Min.</th><th>Typ.</th><th>Max.</th><th>Unit</th><th>Remark</th></tr><tr><td>Is</td><td>LED String Current</td><td>-</td><td>55</td><td>60</td><td>[mA]</td><td>100% duty ratio of LED chip<br/>Note 4-2</td></tr><tr><td>Vs</td><td>LED String Voltage</td><td>39.8</td><td>44.3</td><td>46.5</td><td>[V]</td><td>I=55mA @ 100% duty ratio; Note 4-1, Note 4-3</td></tr><tr><td>ΔVs</td><td>Maximum Vs Voltage Deviation of light bar</td><td>-</td><td>-</td><td>3</td><td>[V]</td><td>I=55mA @ 100% duty ratio; Note 4-2</td></tr><tr><td>P<sub>LED</sub></td><td>LED Light Bar Power Consumption</td><td>-</td><td>9.75</td><td>10.23</td><td>[Watt]</td><td>Note 4-3</td></tr><tr><td>LT<sub>LED</sub></td><td>LED Life Time</td><td>30,000</td><td>-</td><td>-</td><td>[Hour]</td><td>Note 4-4</td></tr><tr><td>OVP</td><td>Over Voltage Protection in system board</td><td>110%<br/>V<sub>max</sub></td><td>-</td><td>-</td><td>[V]</td><td>Note 4-5</td></tr></table> | Symbol                                           | Description | Min.     | Typ.                           | Max.                                | Unit                    | Remark                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Is         | LED String Current             | -              | 55                                | 60         | [mA]     | 100% duty ratio of LED chip<br>Note 4-2 | Vs          | LED String Voltage                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 39.8                 | 44.3            | 46.5                             | [V]      | I=55mA @ 100% duty ratio; Note 4-1, Note 4-3 | ΔVs       | Maximum Vs Voltage Deviation of light bar         | -              | -                                  | 3   | [V]      | I=55mA @ 100% duty ratio; Note 4-2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | P <sub>LED</sub>     | LED Light Bar Power Consumption                   | -              | 9.75                            | 10.23 | [Watt] | Note 4-3 | LT <sub>LED</sub> | LED Life Time                                    | 30,000 | -                                 | -    | [Hour] | Note 4-4 | OVP       | Over Voltage Protection in system board          | 110%<br>V <sub>max</sub> | -                                   | -             | [V] | Note 4-5 |          |                                                  |                |                                   |            |     |   |          |                |                                    |               |     |    |          |                |                                  |            |      |      |          |                |                                     |               |     |     |          |                |                                   |            |          |    |          |     |      |   |    |           |      |      |      |     |          |      |        |      |      |      |    |          |      |           |      |      |      |     |          |  |
| Symbol               | Description                                         | Min.                               | Typ.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 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| Is                   | LED String Current                                  | -                                  | 55                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 60                                                                                                                                                   | [mA]                 | 100% duty ratio of LED chip<br>Note 4-2      |            |        |       |        |                                    |                                                     |      |         |      |                      |                                         |                |                                                     |      |                                |      |                         |                                              |            |                                                   |       |                      |        |     |                                    |                  |                                                   |                      |                 |                                  |          |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      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| Vs                   | LED String Voltage                                  | 42                                 | 44.3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 46.5                                                                                                                                                 | [V]                  | I=55mA @ 100% duty ratio; Note 4-1, Note 4-3 |            |        |       |        |                                    |                                                     |      |         |      |                      |                                         |                |                                                     |      |                                |      |                         |                                              |            |                                                   |       |                      |        |     |                                    |                  |                                                   |                      |                 |                                  |          |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      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| ΔVs                  | Maximum Vs Voltage Deviation of light bar           | -                                  | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 3                                                                                                                                                    | [V]                  | I=55mA @ 100% duty ratio; Note 4-2           |            |        |       |        |                                    |                                                     |      |         |      |                      |                                         |                |                                                     |      |                                |      |                         |                                              |            |                                                   |       |                      |        |     |                                    |                  |                                                   |                      |                 |                                  |          |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      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| P <sub>LED</sub>     | LED Light Bar Power Consumption                     | -                                  | 9.75                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 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| LT <sub>LED</sub>    | LED Life Time                                       | 30,000                             | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    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| OVP                  | Over Voltage Protection in system board             | 110%<br>V <sub>max</sub>           | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    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| Vs                   | LED String Voltage                                  | 39.8                               | 44.3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 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| ΔVs                  | Maximum Vs Voltage Deviation of light bar           | -                                  | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    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| P <sub>LED</sub>     | LED Light Bar Power Consumption                     | -                                  | 9.75                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 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| LT <sub>LED</sub>    | LED Life Time                                       | 30,000                             | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    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| OVP                  | Over Voltage Protection in system board             | 110%<br>V <sub>max</sub>           | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    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|                      |                                                     | 34                                 | <div>7. Mechanical Characteristics</div> <div>(Backview Drawing)</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           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        |                |                                   |            |     |   |          |                |                                    |               |     |    |          |                |                                  |            |      |      |          |                |                                     |               |     |     |          |                |                                   |            |          |    |          |     |      |   |    |           |      |      |      |     |          |      |        |      |      |      |    |          |      |           |      |      |      |     |          |  |



# Product Specification

M240UAN02.3

AU OPTRONICS CORPORATION

|                      |                 |                                |             |            |                                    |                      |                 |                                |                         |             |                                    |                              |        |  |       |                         |
|----------------------|-----------------|--------------------------------|-------------|------------|------------------------------------|----------------------|-----------------|--------------------------------|-------------------------|-------------|------------------------------------|------------------------------|--------|--|-------|-------------------------|
| 35                   |                 |                                | Item        |            |                                    |                      | Remark          | Item                           | Specification           |             |                                    |                              | Remark |  |       |                         |
|                      |                 |                                | Q'ty        | Dimension  |                                    | Weight(kg)           |                 |                                | Q'ty                    | Dimension   |                                    | Weight(kg)                   |        |  |       |                         |
|                      |                 |                                |             |            |                                    |                      |                 |                                |                         |             |                                    |                              |        |  |       |                         |
|                      |                 |                                |             |            |                                    |                      |                 |                                |                         |             |                                    |                              |        |  |       |                         |
|                      |                 |                                | Panel       | 1          | 528.2(H) × 342.4(V) × 12.5(D) Typ. |                      | 2.4             |                                | Panel                   | 1           | 528.2(H) × 342.4(V) × 12.5(D) Typ. |                              | 2.4    |  |       |                         |
|                      |                 |                                | Cushion     | 1          | -                                  |                      | 0.6             |                                | Cushion                 | 1           | -                                  |                              | 0.72   |  |       |                         |
|                      |                 |                                | Box         | 1          | 619(L)mm x 373(W)mm x42(H)mm       |                      | 1.5             |                                | without Panel & cushion | Box         | 1                                  | 617(L)mm x 362(W)mm x42(H)mm |        |  | 1.39  | without Panel & cushion |
|                      |                 |                                | Packing Box | 11 pcs/Box | 619(L)mm x 373(W)mm x42(H)mm       |                      | 28.5            |                                | with panel & cushion    | Packing Box | 11 pcs/Box                         | 617(L)mm x 362(W)mm x42(H)mm |        |  | 28.51 | with panel & cushion    |
| Pallet               | 1               | 1315(L)mm x1150(W)mm x132(H)mm |             | 17.00      |                                    | Pallet               | 1               | 1315(L)mm x1150(W)mm x138(H)mm |                         | 13.75       |                                    |                              |        |  |       |                         |
| Pallet after Packing | 18 boxes/pallet | 1315(L)mm x1150(W)mm x132(H)mm |             | 530        |                                    | Pallet after Packing | 18 boxes/pallet | 1315(L)mm x1150(W)mm x138(H)mm |                         | 526.93      |                                    |                              |        |  |       |                         |

## I Handling Precautions

- 1) Since polarizer is easily damaged, do not touch or press the surface of polarizer with hand.
- 2) Be sure to turn off power supply when inserting or disconnecting from input connector.
- 3) Wipe off water drop immediately. Long contact with water may cause discoloration or spots.
- 4) When the panel surface is soiled, wipe it with absorbent cotton or other soft cloth.
- 5) Since the panel is made of glass, it may break or crack if dropped or bumped on hard surface.
- 6) Since CMOS LSI is used in this module, take care of static electricity and insure human earth when handling.
- 7) Do not open or modify the Module Assembly.
- 8) Do not press the reflector sheet at the back of the module to any directions.
- 9) In case a TFT-LCD Module has to be put back into the packing container slot after once it was taken out from the container, do not press the center of the LED lightbar edge. Otherwise the TFT-LCD Module may be damaged.
- 10) Insert or pull out the interface connector, be sure not to rotate nor tilt it of the TFT-LCD Module.
- 11) Do not twist nor bend the TFT -LCD Module even momentary. It should be taken into consideration that no bending/twisting forces are applied to the TFT-LCD Module from outside. Otherwise the TFT-LCD Module may be damaged.
- 12) Please avoid touching COF position while you are doing mechanical design.
- 13) When storing modules as spares for a long time, the following precaution is necessary: Store them in a dark place. Do not expose the module to sunlight or fluorescent light. Keep the temperature between 5°C and 35°C at normal humidity.
- 14) Do not apply the same pattern for a long time, it will enhance relevant defect.



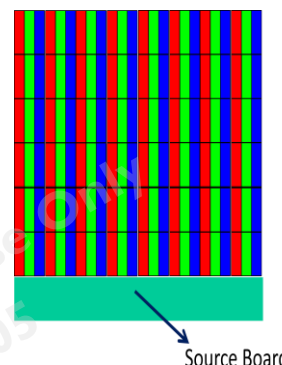
## 2 General Description

This specification applies to the 24 inch wide Color a-Si TFT-LCD Module M240UAN02.0. The display supports the WUXGA - 1920(H) x 1200(V) screen format and 16.7M colors (8 bits data input). The input interface is Dual channel LVDS and this module doesn't contain an driver board for backlight.

### 2.1 Display Characteristics

The following items are characteristics summary on the table under 25°C condition:

| ITEMS                                                | Unit                 | SPECIFICATIONS                                                                                                                   |
|------------------------------------------------------|----------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Screen Diagonal                                      | [mm]                 | 611.32 (24.07")                                                                                                                  |
| Active Area                                          | [mm]                 | 518.4 (H) x 324.0 (V)                                                                                                            |
| Pixels H x V                                         | -                    | 1920(x3) x 1200                                                                                                                  |
| Pixel Pitch                                          | [um]                 | 270 (per one triad) x270                                                                                                         |
| Pixel Arrangement                                    | -                    | R.G.B. Vertical Stripe <b>Note 2-1</b>                                                                                           |
| Display Mode                                         | -                    | AHVA Mode (Advanced Hyper-Viewing Angle), Normally Black                                                                         |
| White Luminance ( Center )                           | [cd/m <sup>2</sup> ] | 250 (Typ.)                                                                                                                       |
| Contrast Ratio                                       | -                    | 1000 (Typ.)                                                                                                                      |
| Response Time                                        | [msec]               | 14 (Typ., GTG)                                                                                                                   |
| Power Consumption<br>(LCD Module + Backligh unit)    | [Watt]               | 14.65(Typ.)<br>LCD module : PDD (Typ.)=4.9W @ White pattern,Fv=60 Hz<br>Backlight unit : P <sub>BLU</sub> (Typ.) =9.75W @Is=55mA |
| Weight                                               | [Grams]              | 2,400(Typ.) / 2,520(Max.)                                                                                                        |
| Outline Dimension                                    | [mm]                 | 528.2(H) x 342.4(V) x 12.2(D) Typ.                                                                                               |
| Electrical Interface                                 | -                    | Dual channel LVDS                                                                                                                |
| Support Color                                        | -                    | 16.7M colors (8bits RGB data input)                                                                                              |
| Surface Treatment                                    | -                    | Anti-Glare, AG25                                                                                                                 |
| Temperature Range<br>Operating<br>Storage (Shipping) | [°C]<br>[°C]         | 0 to +50<br>-20 to +60                                                                                                           |
| RoHS Compliance                                      | -                    | RoHS Compliance                                                                                                                  |
| TCO Compliance                                       | -                    | TCO 8.0 Compliance                                                                                                               |
| Low Blue Light Compliance                            | -                    | LED blue chip wavelength > 452nm                                                                                                 |



**Note 2-1:** The following shows the figure of pixel arrangement

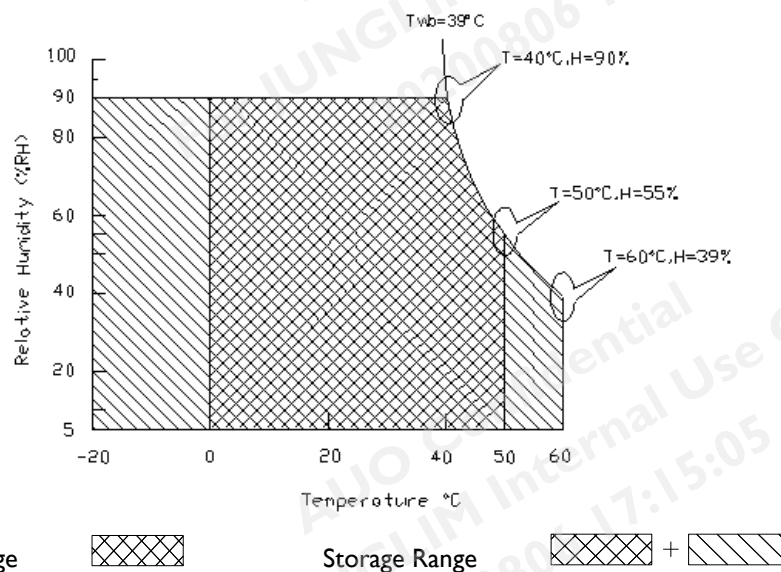
## 2.2 Absolute Maximum Rating of Environment

Permanent damage may occur if exceeding the following maximum rating.

| Symbol | Description                           | Min. | Max. | Unit  | Remark                                  |
|--------|---------------------------------------|------|------|-------|-----------------------------------------|
| TOP    | Operating Temperature                 | 0    | +50  | [°C]  | <b>Note 2-1</b>                         |
| TGS    | Glass surface temperature (operation) | 0    | +65  | [°C]  | <b>Note 2-1</b><br>Function judged only |
| HOP    | Operation Humidity                    | 5    | 90   | [%RH] | <b>Note 2-1</b>                         |
| TST    | Storage Temperature                   | -20  | +60  | [°C]  |                                         |
| HST    | Storage Humidity                      | 5    | 90   | [%RH] |                                         |

**Note 2-2:** Temperature and relative humidity range are shown as the below figure.

1. 90% RH Max (  $T_a \leq 39^\circ\text{C}$  )
2. Max wet-bulb temperature at  $39^\circ\text{C}$  or less. (  $T_a \leq 39^\circ\text{C}$  )
3. No condensation





## 2.3 Optical Characteristics

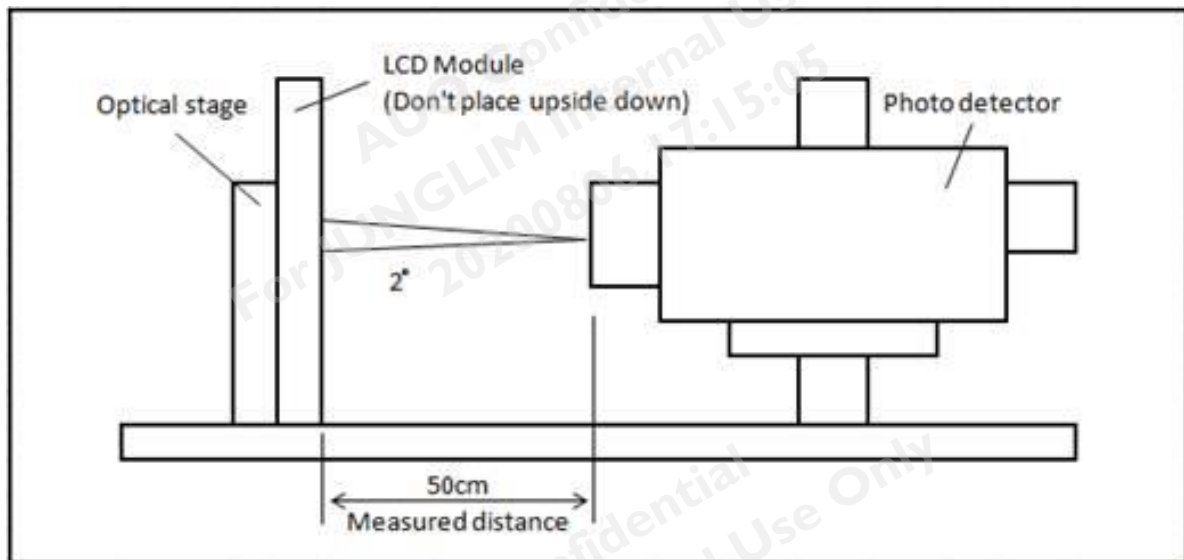
The optical characteristics are measured on the following test condition.

### Test Condition:

1. Equipment setup: Please refer to **Note 2-2**.
2. Panel Lighting time: 30 minutes
3. VDD=5.0V, Fv=60Hz, Is=55mA, Ta=25°C
4. L/B must be in the bottom side at normal and MNT aging state

| Symbol              | Description                         |              | Min.  | Typ.  | Max.  | Unit     | Remark                        |
|---------------------|-------------------------------------|--------------|-------|-------|-------|----------|-------------------------------|
| L <sub>w</sub>      | White Luminance (Center of screen)  |              | 200   | 250   | -     | [cd/m2]  | <b>Note 2-2</b><br>By SR-3    |
| L <sub>uni</sub>    | Luminance Uniformity (9 points)     |              | 75    | 80    | -     | [%]      | <b>Note 2-3</b><br>By SR-3    |
| CR                  | Contrast Ratio (Center of screen)   |              | 700   | 1000  | -     | -        | <b>Note 2-4</b><br>By SR-3    |
| θ <sub>R</sub>      | Horizontal Viewing Angle<br>(CR=10) | Right        | 85    | 89    | -     | [degree] | <b>Note 2-5</b><br>By SR-3    |
| θ <sub>L</sub>      |                                     | Left         | 85    | 89    | -     |          |                               |
| Φ <sub>H</sub>      | Vertical Viewing Angle<br>(CR=10)   | Up           | 85    | 89    | -     |          |                               |
| Φ <sub>L</sub>      |                                     | Down         | 85    | 89    | -     |          |                               |
| θ <sub>R</sub>      | Horizontal Viewing Angle<br>(CR=5)  | Right        | 85    | 89    | -     |          |                               |
| θ <sub>L</sub>      |                                     | Left         | 85    | 89    | -     |          |                               |
| Φ <sub>H</sub>      | Vertical Viewing Angle<br>(CR=5)    | Up           | 85    | 89    | -     |          |                               |
| Φ <sub>L</sub>      |                                     | Down         | 85    | 89    | -     |          |                               |
| T <sub>GTG</sub>    | Response Time                       | Gray To Gray | -     | 14    | -     | [msec]   | <b>Note 2-6</b><br>By TRD-100 |
| R <sub>x</sub>      | Color Coordinates<br>(CIE 1931)     | Red x        | 0.627 | 0.657 | 0.687 | -        | By SR-3                       |
| R <sub>y</sub>      |                                     | Red y        | 0.303 | 0.333 | 0.363 |          |                               |
| G <sub>x</sub>      |                                     | Green x      | 0.270 | 0.300 | 0.330 |          |                               |
| G <sub>y</sub>      |                                     | Green y      | 0.611 | 0.641 | 0.671 |          |                               |
| B <sub>x</sub>      |                                     | Blue x       | 0.114 | 0.144 | 0.174 |          |                               |
| B <sub>y</sub>      |                                     | Blue y       | 0.020 | 0.050 | 0.080 |          |                               |
| W <sub>x</sub>      |                                     | White x      | 0.283 | 0.313 | 0.343 |          |                               |
| W <sub>y</sub>      |                                     | White y      | 0.299 | 0.329 | 0.359 |          |                               |
| sRGB coverage ratio |                                     |              |       | 99    |       | [%]      | By SR-3                       |

**Note 2-2:** Equipment setup :

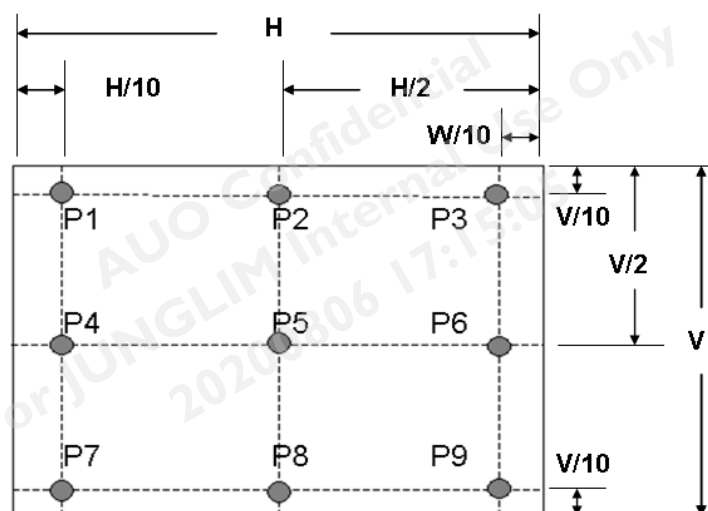


**Note 2-3:** Luminance Uniformity Measurement

**Definition:**

$$\text{Luminance Uniformity} = \frac{\text{Minimum Luminance of 9 Points (P1 ~ P9)}}{\text{Maximum Luminance of 9 Points (P1 ~ P9)}}$$

*a. Test pattern: White Pattern*



**Note 2-4:** Contrast Ratio Measurement

**Definition:**

$$\text{Contrast Ratio} = \frac{\text{Luminance of White pattern}}{\text{Luminance of Black pattern}}$$

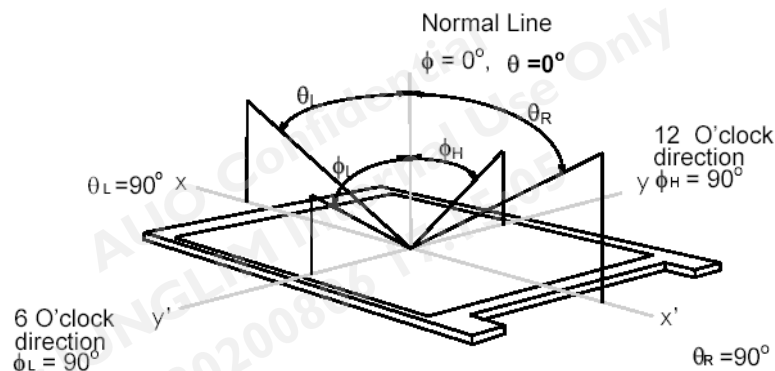
- a. Measured position: Center of screen (P5) & perpendicular to the screen ( $\theta=\Phi=0^\circ$ )

**Note 2-5:** Viewing angle measurement

**Definition:** The angle at which the contrast ratio is greater than 10 & 5 .

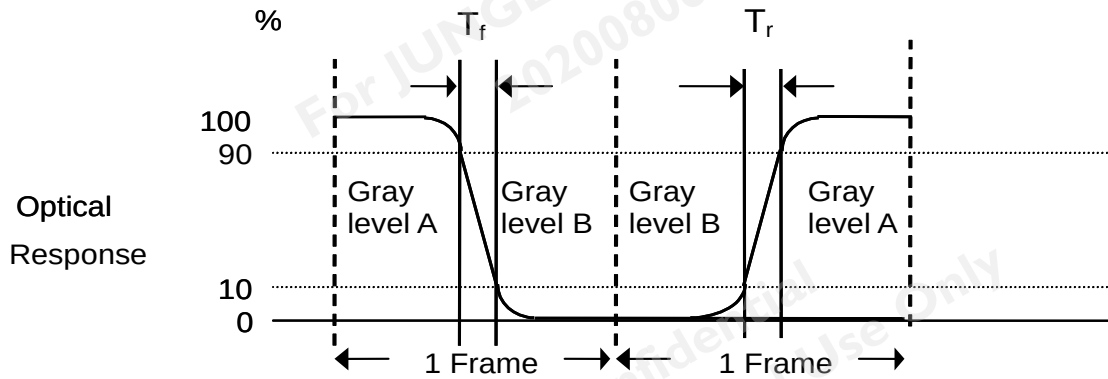
- a. Horizontal view angle: Divide to left & right ( $\theta_L$  &  $\theta_R$ )

Vertical view angle: Divide to up & down ( $\Phi_H$  &  $\Phi_L$ )



## Note 2-6: Response time measurement

The output signals of photo detector are measured when the input signals are changed from “Gray level A” to “Gray level B” (falling time,  $T_f$ ), and from “Gray level B” to “Gray level A” (rising time,  $T_r$ ), respectively. The response time is interval between the 10% and 90% of optical response.



The gray to gray response time is defined as the following table.

| Gray Level to Gray Level |      | Target gray level |     |      |      |      |
|--------------------------|------|-------------------|-----|------|------|------|
|                          |      | L0                | L63 | L127 | L191 | L255 |
| Start gray level         | L0   |                   |     |      |      |      |
|                          | L63  |                   |     |      |      |      |
|                          | L127 |                   |     |      |      |      |
|                          | L191 |                   |     |      |      |      |
|                          | L255 |                   |     |      |      |      |

■  $T_{GTG\_typ}$  is the total average time at rising time and falling time of gray to gray.

## 2.4 Mechanical Characteristics

| Symbol   | Description          | Min. | Max. | Unit  | Remark          |
|----------|----------------------|------|------|-------|-----------------|
| $P_{bc}$ | Backside Compression | 2.5  | -    | [Kgf] | <b>Note 2-9</b> |

### Note 2-9: 1. Test condition:

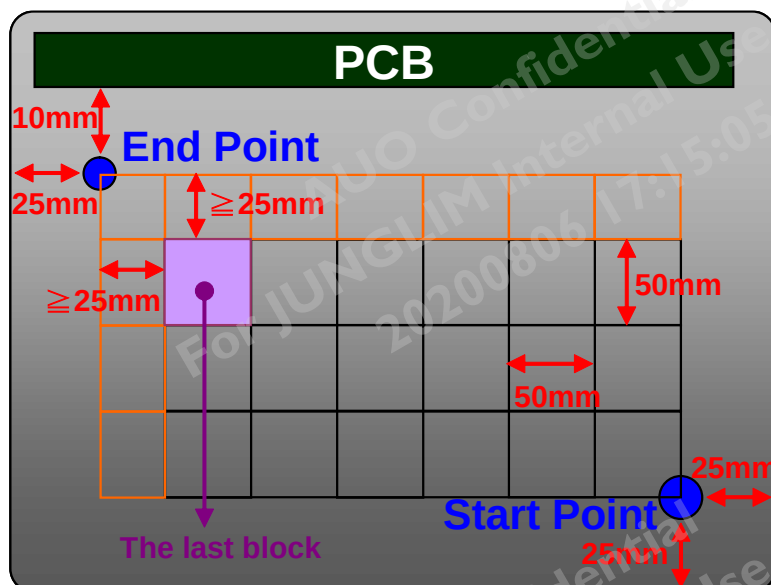
- (1) Ambient illumination : 10 ~ 15 lux
- (2) Test pattern : Gray (L128)
- (3) Distance of judgment : 30 cm from surface of LCD panel
- (4) Viewing angle of judgment : Front
- (5) Test jig : Push-Pull Gauge & Flat Head ( $\Phi$  2)

### 2. Test Method:

The point is at a distance from right-downside 25mm x 25mm defined as the Start Point of Measure Points, and the point is at a distance 25mm from left-side & around 10mm from PCB defined as the End Point.

Align 50mm x 50mm block from Start Point on the Bezel Back, and the corners of each block are Measure Points.

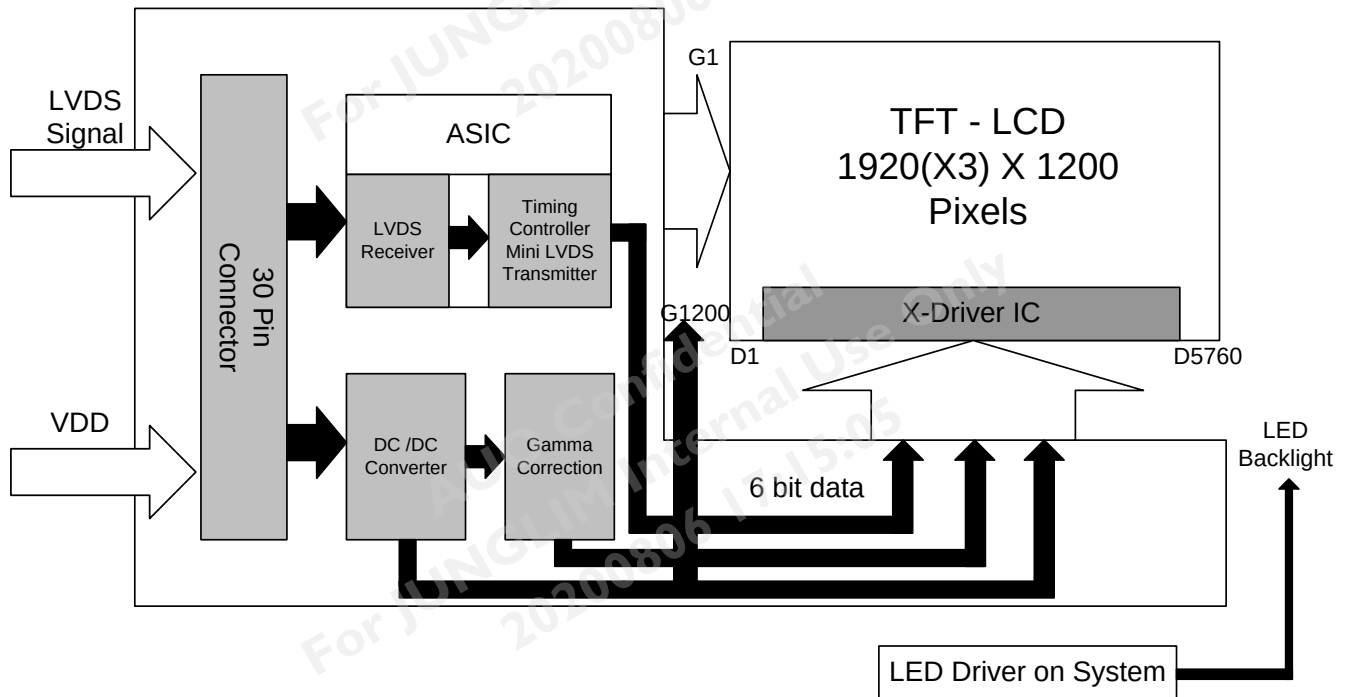
If the distance from the last block to each side of the End Point  $\geq$  25mm, add other blocks to make sure that most area of Bezel Back can be measured.



## 3 TFT-LCD Module

### 3.1 Block Diagram

The following shows the block diagram of the 24 inch Color TFT-LCD Module.





## 3.2 Interface Connection

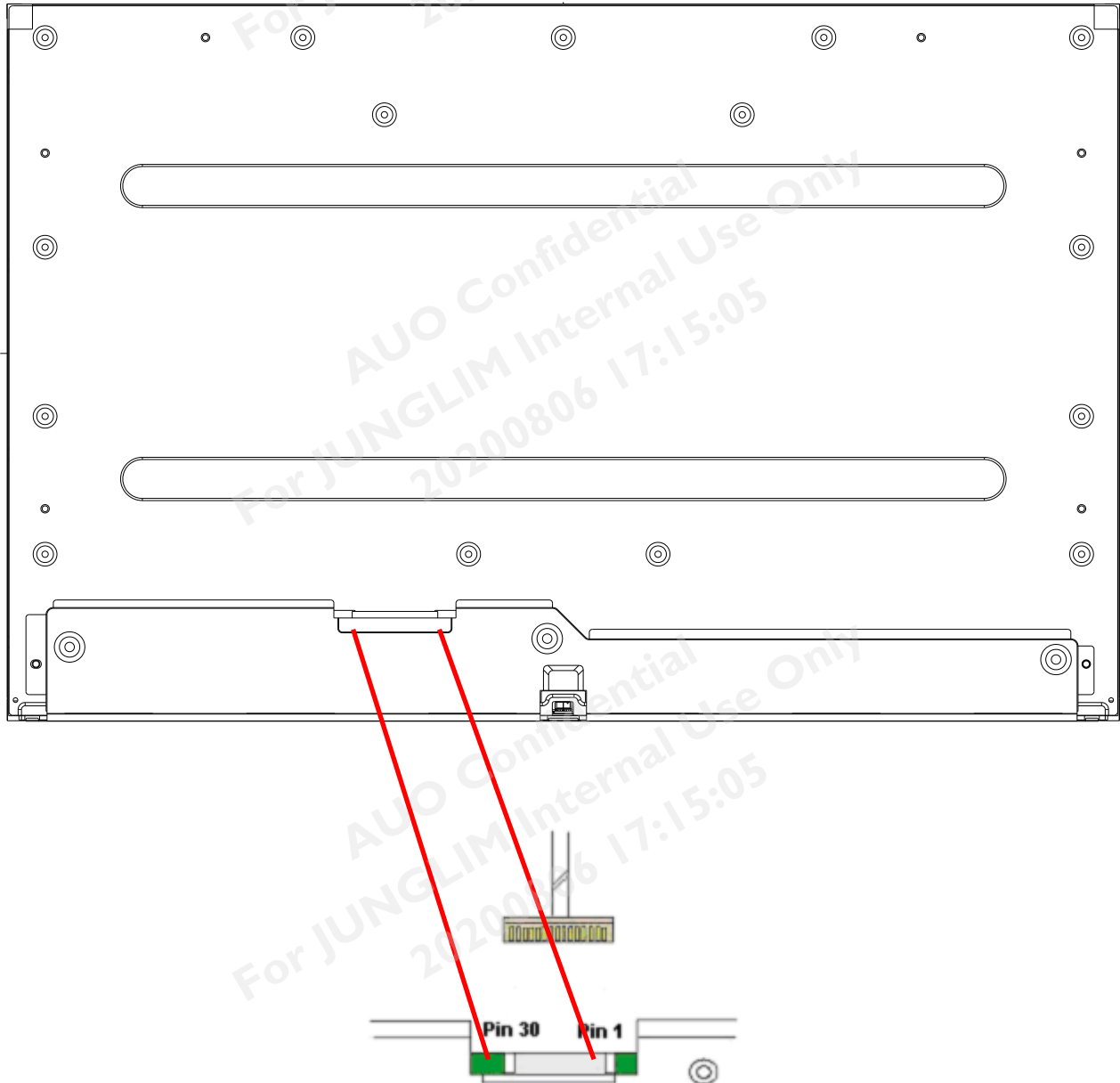
### 3.2.1 Connector Type

|                   |              |                        |                |
|-------------------|--------------|------------------------|----------------|
| TFT-LCD Connector | Manufacturer | P-TWO                  | STM            |
|                   | Part Number  | I87034-3009            | MSBKT2407P30HB |
| Mating Connector  | Manufacturer | JAE or Compatible      |                |
|                   | Part Number  | FI-X30HL (Locked Type) |                |

### 3.2.2 Connector Pin Assignment

| PIN # | Symbol  | Description                                         | Remark |
|-------|---------|-----------------------------------------------------|--------|
| 1     | RxO0-   | Negative LVDS differential data input (Odd data)    |        |
| 2     | RxO0+   | Positive LVDS differential data input (Odd data)    |        |
| 3     | RxO1-   | Negative LVDS differential data input (Odd data)    |        |
| 4     | RxO1+   | Positive LVDS differential data input (Odd data)    |        |
| 5     | RxO2-   | Negative LVDS differential data input (Odd data)    |        |
| 6     | RxO2+   | Positive LVDS differential data input (Odd data)    |        |
| 7     | GND     | Ground                                              |        |
| 8     | RxOCLK- | Negative LVDS differential clock input (Odd clock)  |        |
| 9     | RxOCLK+ | Positive LVDS differential clock input (Odd clock)  |        |
| 10    | RxO3-   | Negative LVDS differential data input (Odd data)    |        |
| 11    | RxO3+   | Positive LVDS differential data input (Odd data)    |        |
| 12    | RxE0-   | Negative LVDS differential data input (Even data)   |        |
| 13    | RxE0+   | Positive LVDS differential data input (Even data)   |        |
| 14    | GND     | Ground                                              |        |
| 15    | RxE1-   | Negative LVDS differential data input (Even data)   |        |
| 16    | RxE1+   | Positive LVDS differential data input (Even data)   |        |
| 17    | GND     | Ground                                              |        |
| 18    | RxE2-   | Negative LVDS differential data input (Even data)   |        |
| 19    | RxE2+   | Positive LVDS differential data input (Even data)   |        |
| 20    | RxECLK- | Negative LVDS differential clock input (Even clock) |        |
| 21    | RxECLK+ | Positive LVDS differential clock input (Even clock) |        |
| 22    | RxE3-   | Negative LVDS differential data input (Even data)   |        |
| 23    | RxE3+   | Positive LVDS differential data input (Even data)   |        |
| 24    | NC      | No connection (for AUO test only. Do not connect)   |        |
| 25    | NC      | No connection (for AUO test only. Do not connect)   |        |
| 26    | NC      | No connection (for AUO test only. Do not connect)   |        |

|    |     |                                                   |  |
|----|-----|---------------------------------------------------|--|
| 27 | NC  | No connection (for AUO test only. Do not connect) |  |
| 28 | VDD | Power Supply Input Voltage                        |  |
| 29 | VDD | Power Supply Input Voltage                        |  |
| 30 | VDD | Power Supply Input Voltage                        |  |



## 3.3 Electrical Characteristics

### 3.3.1 Absolute Maximum Rating

Permanent damage may occur if exceeding the following maximum rating.

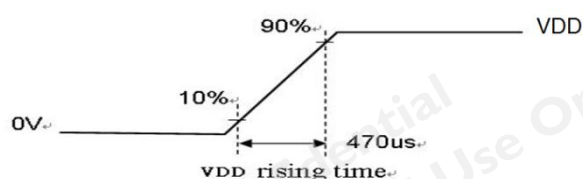
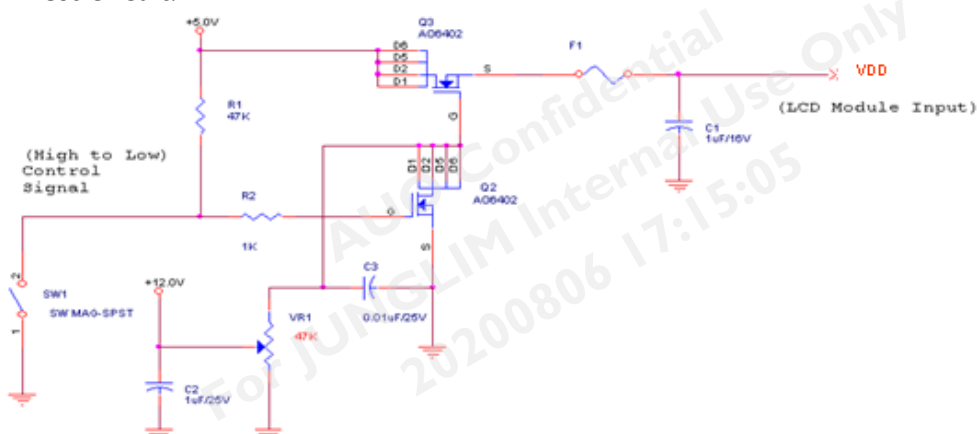
| Symbol | Description                | Min     | Max | Unit   | Remark  |
|--------|----------------------------|---------|-----|--------|---------|
| VDD    | Power Supply Input Voltage | GND-0.3 | 6.0 | [Volt] | Ta=25°C |

### 3.3.2 Recommended Operating Condition

| Symbol | Description                      | Min | Typ  | Max  | Unit   | Remark                            |
|--------|----------------------------------|-----|------|------|--------|-----------------------------------|
| VDD    | Power supply Input voltage       | 4.5 | 5.0  | 5.5  | [Volt] |                                   |
| IDD    | Power supply Input Current (RMS) | -   | 0.98 | 1.18 | [A]    | VDD= 5.0V, White Pattern, Fv=60Hz |
|        |                                  |     | 1.22 | 1.46 | [A]    | VDD= 5.0V, White Pattern, Fv=75Hz |
| PDD    | VDD Power Consumption            | -   | 4.90 | 5.90 | [Watt] | VDD= 5.0V, White Pattern, Fv=60Hz |
|        |                                  |     | 6.10 | 7.30 | [Watt] | VDD= 5.0V, White Pattern, Fv=75Hz |
| IRush  | Inrush Current                   | -   | -    | 3.0  | [A]    | <b>Note 3-1</b>                   |
| VDDrp  | Allowable VDD Ripple Voltage     | -   | -    | 500  | [mV]   | VDD= 5.0V, White Pattern, Fv=75Hz |

**Note 3-1:** Inrush Current measurement:

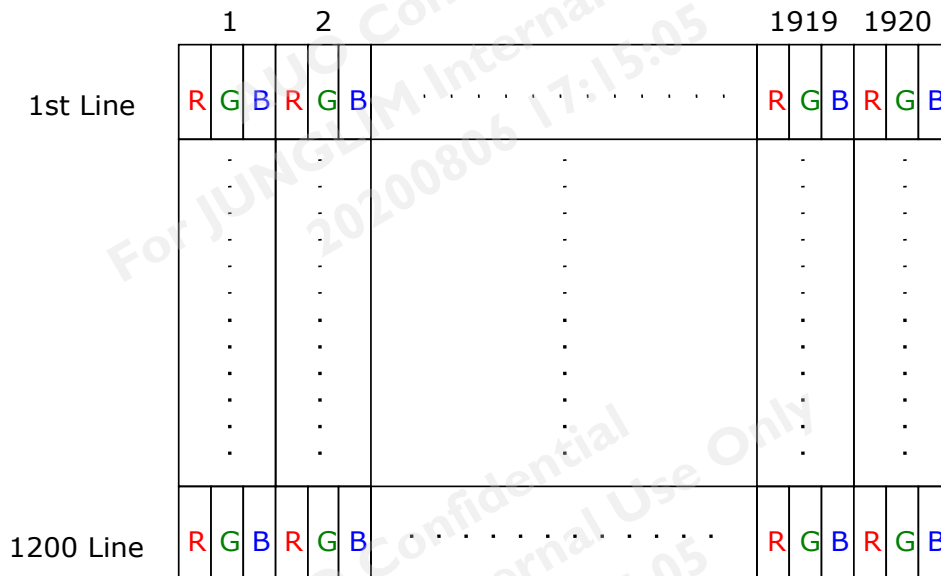
Test circuit:



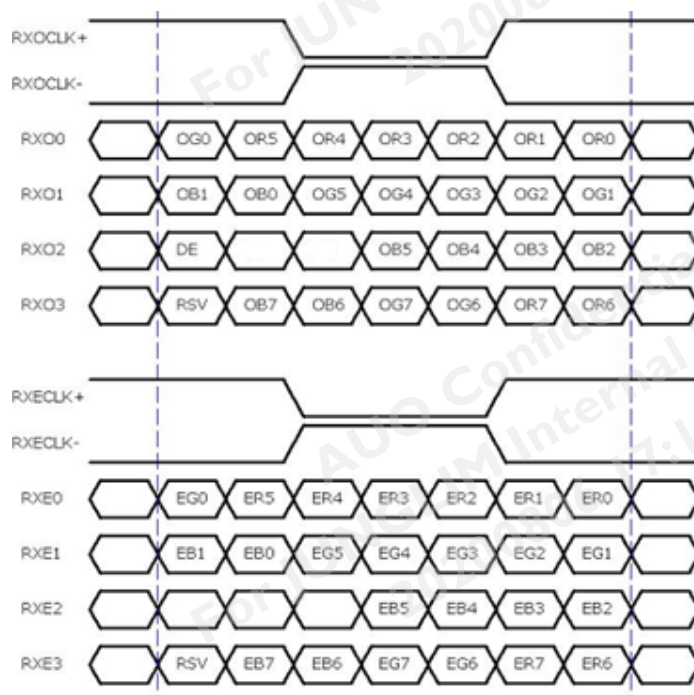
The duration of VDD rising time: 470us.

## 3.4 Signal Characteristics

### 3.4.1 LCD Pixel Format



### 3.4.2 LVDS Data Format



| 8 Bit Color Bit Order |    |    |    |
|-----------------------|----|----|----|
| MSB                   | R7 | G7 | B7 |
|                       | R6 | G6 | B6 |
|                       | R5 | G5 | B5 |
|                       | R4 | G4 | B4 |
|                       | R3 | G3 | B3 |
|                       | R2 | G2 | B2 |
|                       | R1 | G1 | B1 |
| LSB                   | R0 | G0 | B0 |

#### Note 3-2:

- O = "Odd Pixel Data" E = "Even Pixel Data"
- Refer to 3.4.1 LCD pixel format, the 1st data is 1 (Odd Pixel Data), the 2<sup>nd</sup> data is 2 (Even Pixel Data) and the last data is 1920 (Even Pixel Data).

## 3.4.3 Color versus Input Data

The following table is for color versus input data (8bit). The higher the gray level, the brighter the color.

| Color    | Gray Level | Color Input Data                              |    |    |    |    |    |    |    |                                                 |    |    |    |    |    |    |    |                                                |    |    |    |    |    |    |       | Remark |
|----------|------------|-----------------------------------------------|----|----|----|----|----|----|----|-------------------------------------------------|----|----|----|----|----|----|----|------------------------------------------------|----|----|----|----|----|----|-------|--------|
|          |            | RED data<br>( <b>MSB</b> :R7, <b>LSB</b> :R0) |    |    |    |    |    |    |    | GREEN data<br>( <b>MSB</b> :G7, <b>LSB</b> :G0) |    |    |    |    |    |    |    | BLUE data<br>( <b>MSB</b> :B7, <b>LSB</b> :B0) |    |    |    |    |    |    |       |        |
|          |            | R7                                            | R6 | R5 | R4 | R3 | R2 | R1 | R0 | G7                                              | G6 | G5 | G4 | G3 | G2 | G1 | G0 | B7                                             | B6 | B5 | B4 | B3 | B2 | B1 | B0    |        |
| Black    | -          | 0                                             | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0                                               | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0                                              | 0  | 0  | 0  | 0  | 0  | 0  |       |        |
| White    | -          | 1                                             | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 1                                               | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 1                                              | 1  | 1  | 1  | 1  | 1  | 1  |       |        |
| Gray 127 | -          | 0                                             | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 0                                               | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 0                                              | 1  | 1  | 1  | 1  | 1  | 1  |       |        |
| Red      | L0         | 0                                             | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0                                               | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0                                              | 0  | 0  | 0  | 0  | 0  | 0  | Black |        |
|          | :          | :                                             | :  | :  | :  | :  | :  | :  | :  | :                                               | :  | :  | :  | :  | :  | :  | :  | :                                              | :  | :  | :  | :  | :  | :  |       |        |
|          | L255       | 1                                             | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 0                                               | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0                                              | 0  | 0  | 0  | 0  | 0  | 0  |       |        |
| Green    | L0         | 0                                             | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0                                               | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0                                              | 0  | 0  | 0  | 0  | 0  | 0  | Black |        |
|          | :          | :                                             | :  | :  | :  | :  | :  | :  | :  | :                                               | :  | :  | :  | :  | :  | :  | :  | :                                              | :  | :  | :  | :  | :  | :  |       |        |
|          | L255       | 0                                             | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 1                                               | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 0                                              | 0  | 0  | 0  | 0  | 0  | 0  |       |        |
| Blue     | L0         | 0                                             | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0                                               | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0                                              | 0  | 0  | 0  | 0  | 0  | 0  | Black |        |
|          | :          | :                                             | :  | :  | :  | :  | :  | :  | :  | :                                               | :  | :  | :  | :  | :  | :  | :  | :                                              | :  | :  | :  | :  | :  | :  |       |        |
|          | L255       | 0                                             | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0                                               | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 1                                              | 1  | 1  | 1  | 1  | 1  | 1  |       |        |

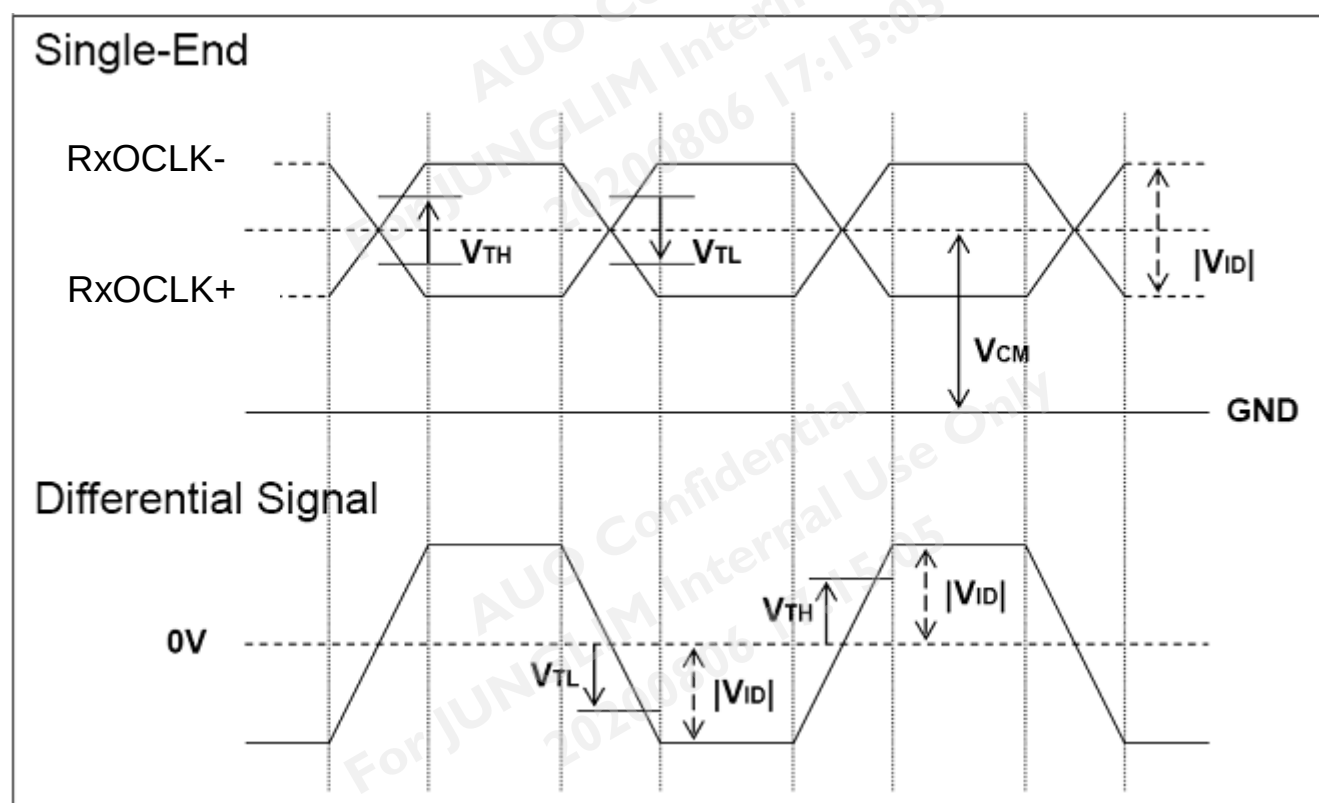
## 3.4.4 LVDS Specification

### a. DC Characteristics:

| Symbol     | Description                            | Min  | Typ  | Max  | Units | Condition               |
|------------|----------------------------------------|------|------|------|-------|-------------------------|
| $V_{TH}$   | LVDS Differential Input High Threshold | -    | -    | +100 | [mV]  | $V_{CM} = 1.2V$         |
| $V_{TL}$   | LVDS Differential Input Low Threshold  | -100 | -    | -    | [mV]  | $V_{CM} = 1.2V$         |
| $ V_{ID} $ | LVDS Differential Input Voltage        | 100  | -    | 600  | [mV]  |                         |
| $V_{CM}$   | LVDS Common Mode Voltage               | +1.0 | +1.2 | +1.5 | [V]   | $V_{TH}-V_{TL} = 200mV$ |

### LVDS Signal Waveform:

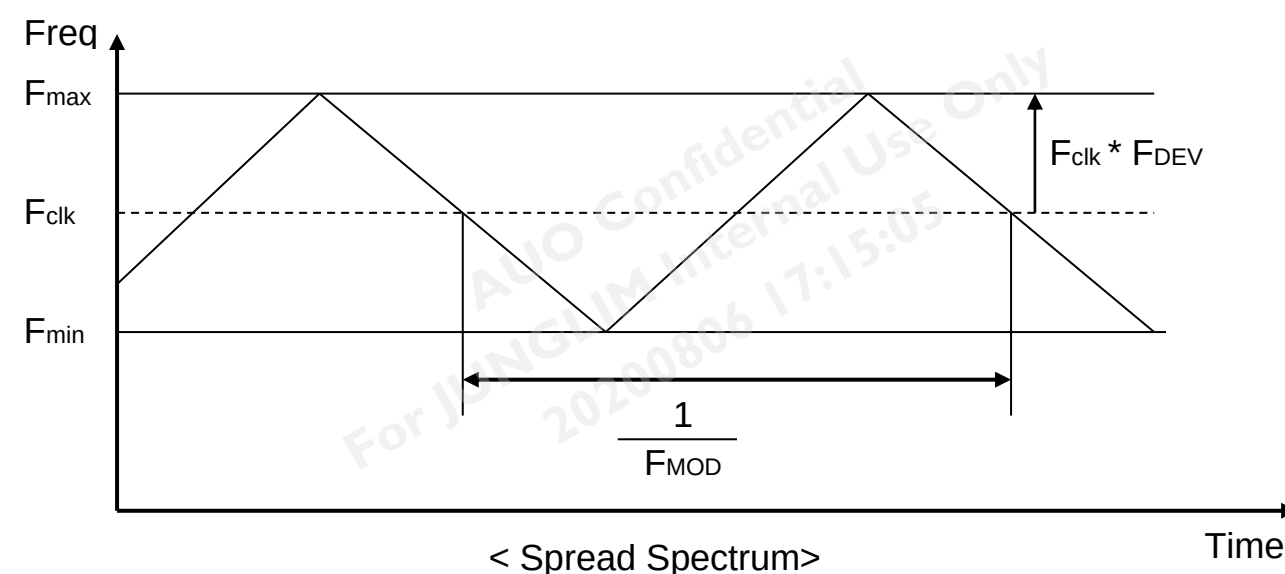
Use RxOCLK- & RxOCLK+ as example.



### b. AC Characteristics:



| Symbol    | Description                                                        | Min | Max     | Unit | Remark |
|-----------|--------------------------------------------------------------------|-----|---------|------|--------|
| $F_{DEV}$ | Maximum deviation of input clock frequency during Spread Spectrum  | -   | $\pm 3$ | %    |        |
| $F_{MOD}$ | Maximum modulation frequency of input clock during Spread Spectrum | -   | 200     | KHz  |        |



Fclk: LVDS Clock Frequency

## 3.4.5 Input Timing Specification

It only support DE mode, and the input timing are shown as the following table.

| Symbol    | Description        |           | Min. | Typ. | Max. | Unit | Remark          |
|-----------|--------------------|-----------|------|------|------|------|-----------------|
| Tv        | Vertical Section   | Period    | 1214 | 1235 | 1965 | Th   |                 |
| Tdisp (v) |                    | Active    | 1200 | 1200 | 1200 | Th   |                 |
| Tblk (v)  |                    | Blanking  | 14   | 35   | 765  | Th   |                 |
| Fv        |                    | Frequency | 49   | 60   | 75   | Hz   |                 |
| Th        | Horizontal Section | Period    | 1000 | 1040 | 1428 | Tclk |                 |
| Tdisp (h) |                    | Active    | 960  | 960  | 960  | Tclk |                 |
| Tblk (h)  |                    | Blanking  | 40   | 80   | 468  | Tclk |                 |
| Fh        |                    | Frequency | 59.5 | 74.0 | 96.3 | KHz  | <b>Note 3-3</b> |
| Tclk      | LVDS Clock         | Period    | 10.4 | 13.0 | 16.8 | ns   | I/Fclk          |
| Fclk      |                    | Frequency | 59.5 | 77.0 | 96.3 | MHz  | <b>Note 3-4</b> |

**Note 3-3:** The equation is listed as following. Please don't exceed the above recommended value.

$$Fh (\text{Min.}) = Fclk (\text{Min.}) / Th (\text{Min.});$$

$$Fh (\text{Typ.}) = Fclk (\text{Typ.}) / Th (\text{Typ.});$$

$$Fh (\text{Max.}) = Fclk (\text{Max.}) / Th (\text{Min.});$$

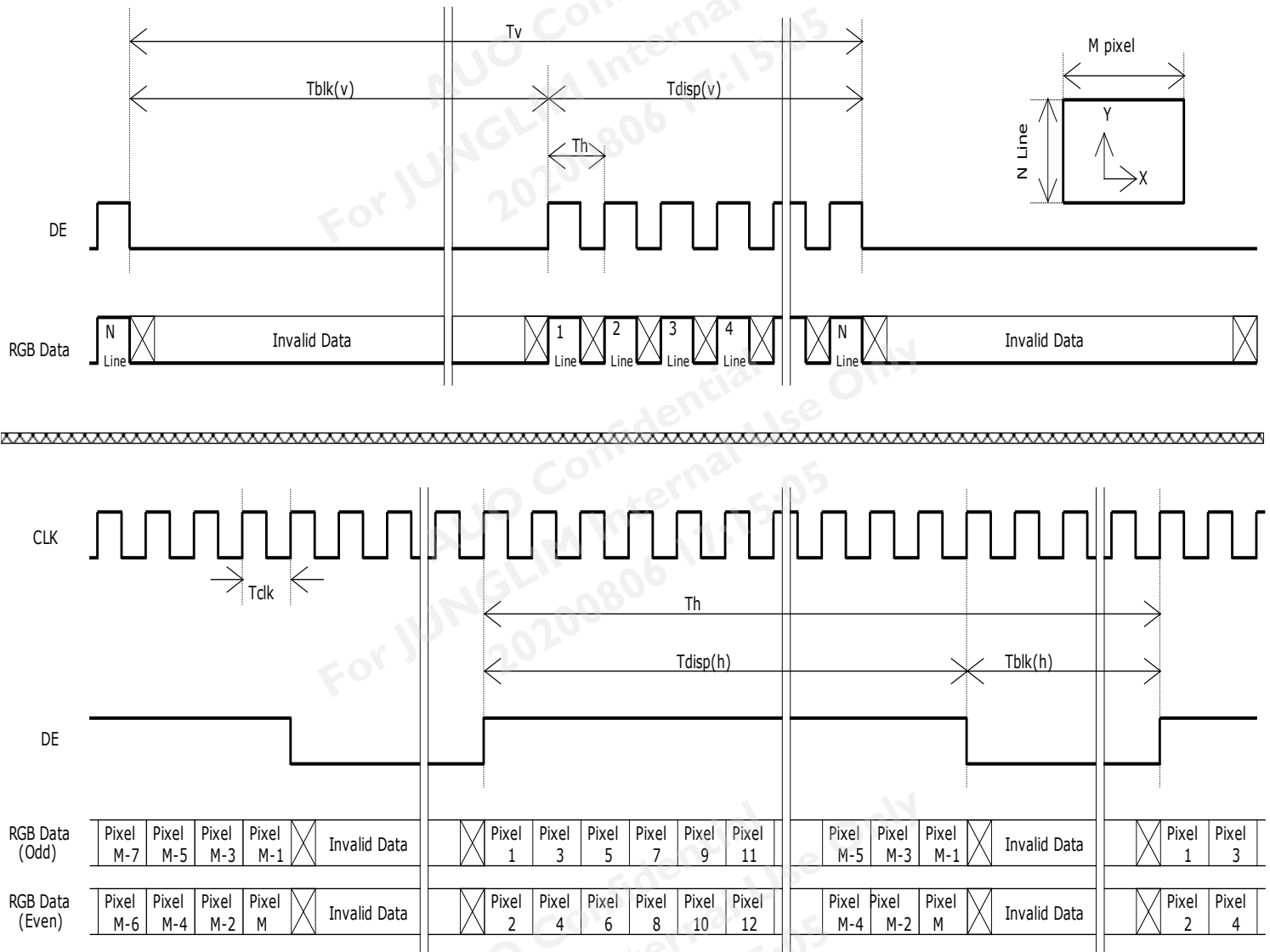
**Note 3-4:** The equation is listed as following. Please don't exceed the above recommended value.

$$Fclk (\text{Min.}) = Fv (\text{Min.}) \times Th (\text{Min.}) \times Tv (\text{Min.});$$

$$Fclk (\text{Typ.}) = Fv (\text{Typ.}) \times Th (\text{Typ.}) \times Tv (\text{Typ.});$$

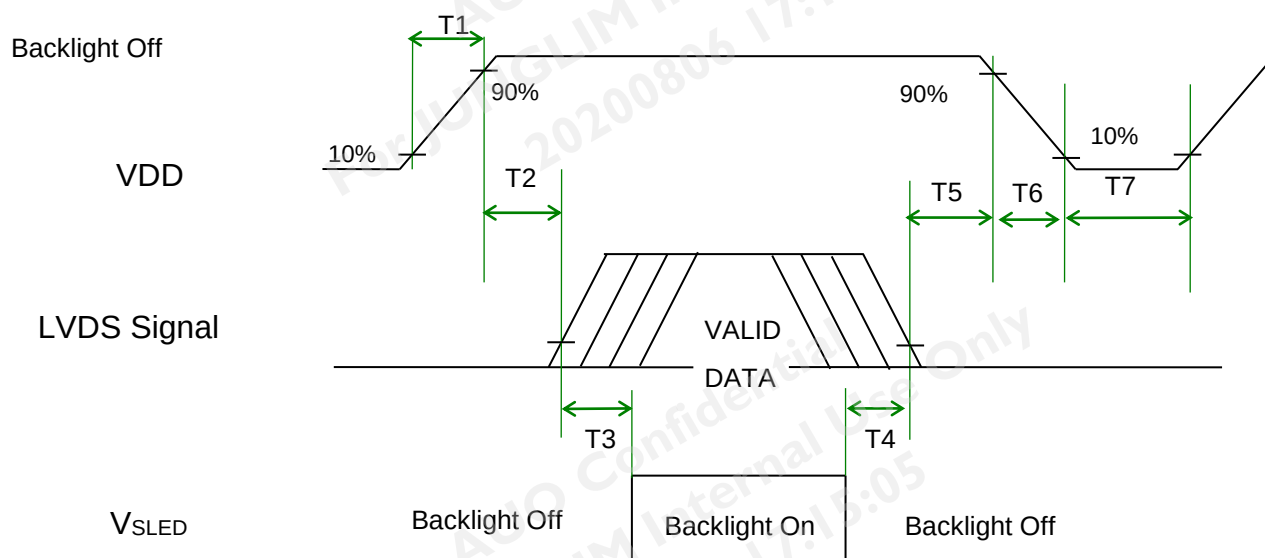
$$Fclk (\text{Max.}) = Fv (\text{Max.}) \times Th (\text{Typ.}) \times Tv (\text{Typ.});$$

## 3.4.6 Input Timing Diagram



## 3.5 Power ON/OFF Sequence

VDD power, LVDS signal and backlight on/off sequence are as following. LVDS signals from any system shall be Hi-Z state when VDD is off.



Power Sequence Timing

| Symbol | Value |      |      | Unit | Remark                             |
|--------|-------|------|------|------|------------------------------------|
|        | Min.  | Typ. | Max. |      |                                    |
| T1     | 0.5   | -    | 10   | [ms] |                                    |
| T2     | 0     | -    | 50   | [ms] |                                    |
| T3     | 500   | -    | -    | [ms] |                                    |
| T4     | 100   | -    | -    | [ms] | <b>Note 3-8</b>                    |
| T5     | 0     | -    | 50   | [ms] | <b>Note 3-5</b><br><b>Note 3-6</b> |
| T6     | 0     | -    | 200  | [ms] | <b>Note 3-6</b><br><b>Note 3-7</b> |
| T7     | 1000  | -    | -    | [ms] |                                    |

**Note 3-5 :** Recommend setting T5 = 0ms to avoid electronic noise when VDD is off.

**Note 3-6 :** During T5 and T6 period , please keep the level of input LVDS signals with Hi-Z state.

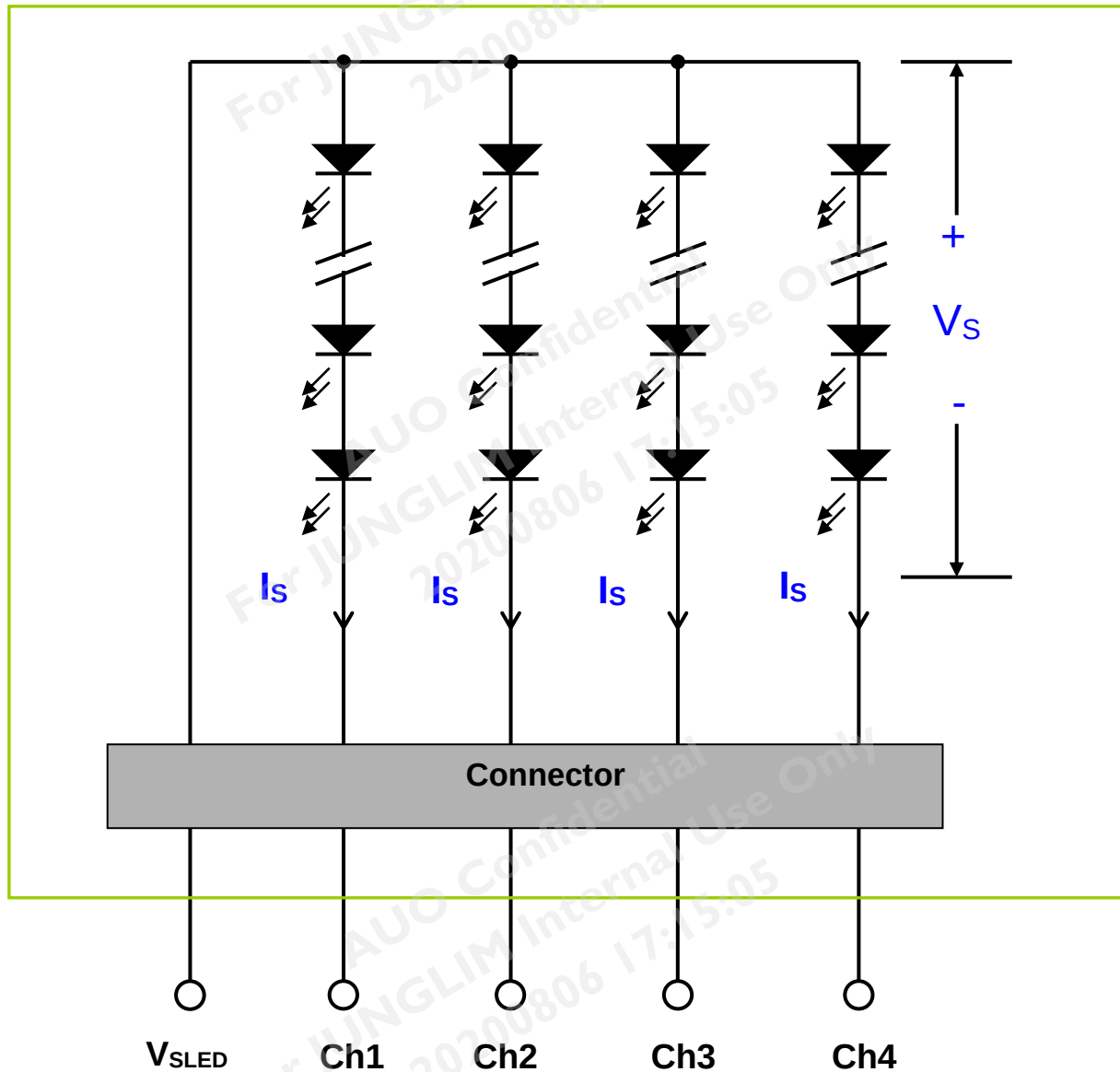
**Note 3-7 :** Voltage of VDD must decay smoothly after power-off. (customer system decide this value)

**Note 3-8 :** If T4 < 100ms, there will be no reliability concern, but the display may momentarily show abnormal screen.

## 4 Backlight Unit

### 4.1 Block Diagram

The following shows the block diagram of the 24 inch Backlight Unit. And it includes 60 pcs LED in the LED light bar. (4 strings and 15 pcs LED of one string).



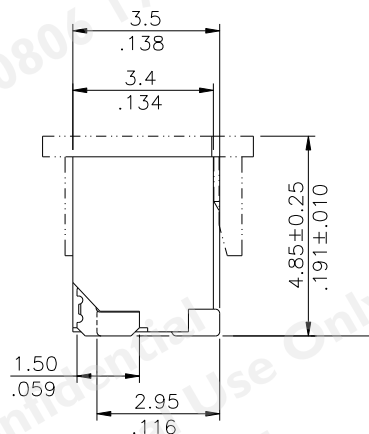
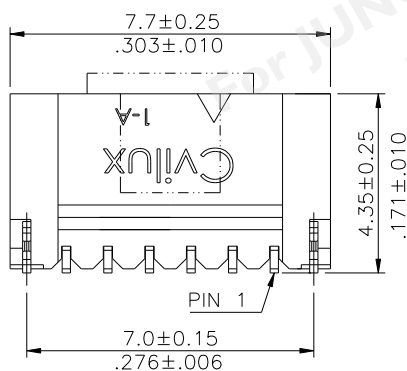
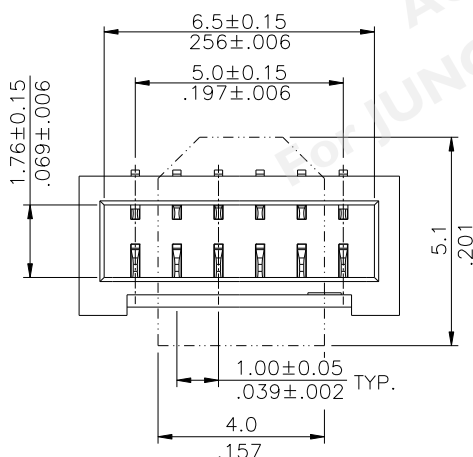
## 4.2 Interface Connection

### 4.2.1 Connector Type

|                     |              |                                                                                                                   |
|---------------------|--------------|-------------------------------------------------------------------------------------------------------------------|
| Backlight Connector | Manufacturer | CVILUX                                                                                                            |
|                     | Part Number  | CII406MIVL0-NH                                                                                                    |
| Mating Connector    | Manufacturer | ENTERY or Compatible                                                                                              |
|                     | Part Number  | HI12K-P06N-00B (Non-Locking type)<br>HI12K-P06N-11B(White) (Locking type)<br>HI12K-P06N-13B(Black) (Locking type) |

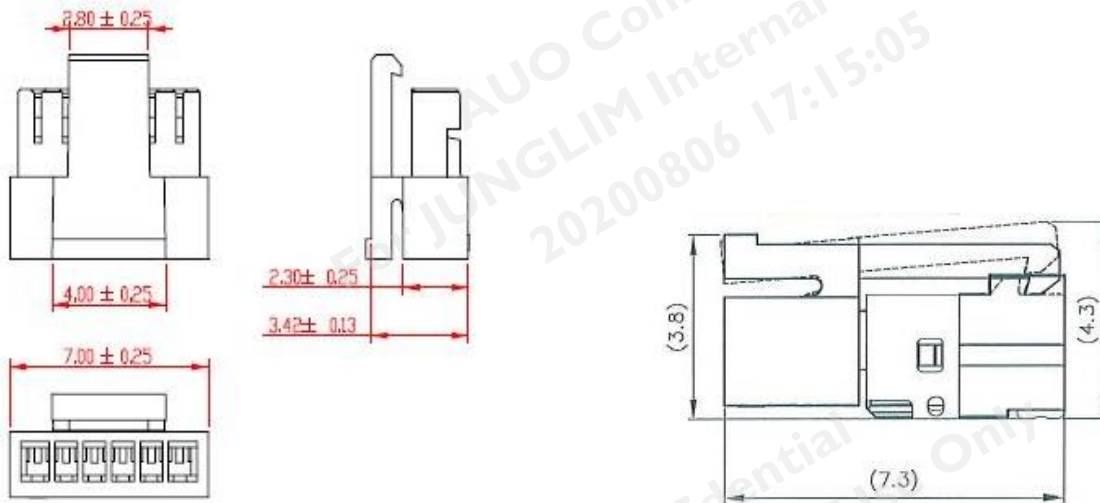
### Backlight Connector dimension:

H×V×D=7.7×3.5×4.35, Pitch=1.0(unit=mm)



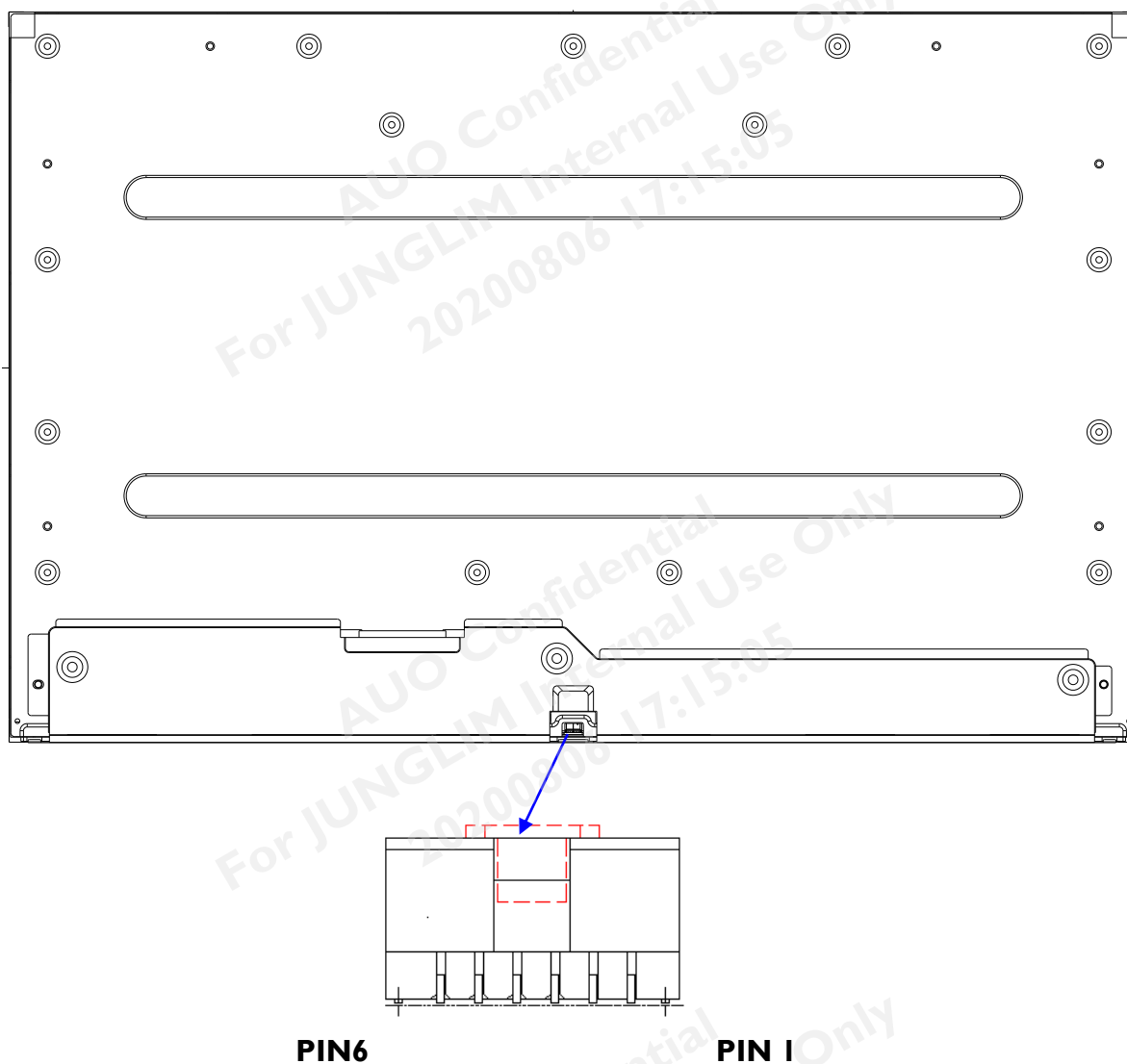


## Mating Connector dimension:



## 4.2.2 Connector Pin Assignment

| Pin# | Symbol            | Description                               | Remark |
|------|-------------------|-------------------------------------------|--------|
| 1    | Ch1               | LED Current Feedback Terminal (Channel 1) |        |
| 2    | Ch2               | LED Current Feedback Terminal (Channel 2) |        |
| 3    | V <sub>SLED</sub> | LED Power Supply Voltage Input Terminal   |        |
| 4    | V <sub>SLED</sub> | LED Power Supply Voltage Input Terminal   |        |
| 5    | Ch3               | LED Current Feedback Terminal (Channel 3) |        |
| 6    | Ch4               | LED Current Feedback Terminal (Channel 4) |        |



## 4.3 Electrical Characteristics

### 4.3.1 Absolute Maximum Rating

Permanent damage may occur if exceeding the following maximum rating.

(Ta=25°C)

| Symbol | Description        | Min | Max | Unit | Remark          |
|--------|--------------------|-----|-----|------|-----------------|
| Is     | LED String Current | 0   | 120 | [mA] | 100% duty ratio |

### 4.3.2 Recommended Operating Condition

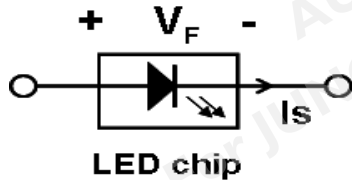
(Ta=25°C)

| Symbol            | Description                               | Min.       | Typ. | Max.  | Unit   | Remark                                               |
|-------------------|-------------------------------------------|------------|------|-------|--------|------------------------------------------------------|
| Is                | LED String Current                        | -          | 55   | 60    | [mA]   | 100% duty ratio of LED chip<br><b>Note 4-7</b>       |
| Vs                | LED String Voltage                        | 39.8       | 44.3 | 46.5  | [Volt] | Is=55mA @ 100% duty ratio; <b>Note 4-1, Note 4-5</b> |
| ΔVs               | Maximum Vs Voltage Deviation of light bar | -          | -    | 3     | [Volt] | Is=55mA @ 100% duty ratio; <b>Note 4-2</b>           |
| P <sub>BLU</sub>  | LED Light Bar Power Consumption           | -          | 9.75 | 10.23 | [Watt] | <b>Note 4-3</b>                                      |
| LT <sub>LED</sub> | LED Life Time                             | 30,000     | -    | -     | [Hour] | <b>Note 4-4</b>                                      |
| OVP               | Over Voltage Protection in system board   | 110% Vsmax | -    | -     | [Volt] | <b>Note 4-5</b>                                      |

**Note 4-1:**  $V_s (\text{Typ.}) = V_F (\text{Typ.}) \times \text{LED No. (one string)}$ ;

a.  $V_F$ : LED chip forward voltage,  $V_F (\text{Min.}) = 2.65\text{V}$ ,  $V_F (\text{Typ.}) = 2.95\text{V}$ ,  $V_F (\text{Max.}) = 3.1\text{V}$

b. The same equation to calculate  $V_s (\text{Min.})$  &  $V_s (\text{Max.})$  for respective  $V_F (\text{Min.})$  &  $V_F (\text{Max.})$ ;



**Note 4-2:**  $\Delta V_s (\text{Max.}) = \Delta V_F \times \text{LED No. (one string)}$ ;

a.  $\Delta V_F$ : LED chip forward voltage deviation; (0.2 V , each Bin of LED  $V_F$ )

**Note 4-3:**  $P_{\text{BLU}} (\text{Typ.}) = V_s (\text{Typ.}) \times I_s (\text{Typ.}) \times 4$  ; ( 4 is total String No. of LED Light bar)

$P_{\text{BLU}} (\text{Max.}) = V_s (\text{Max.}) \times I_s (\text{Typ.}) \times 4$  ;

**Note 4-4:** Definition of life time:

a. Brightness of LED becomes to 50% of its original value

b. Test condition:  $I_s = 55\text{mA}$  and  $25^\circ\text{C}$  (Room Temperature)

**Note 4-5:** Recommendation for LED driver power design:

Due to there are electrical property deviation in LED & monitor set system component after long time operation. AUO strongly recommend the design value of LED driver board OVP (over voltage protection) should be 10% higher than max. value of LED string voltage ( $V_s$ ) at least.

**Note 4-6:** AUO strongly recommend “Analog Dimming” method for backlight brightness control for Wavy Noise Free. Otherwise, recommend that Dimming Control Signal (PWM Signal) should be synchronized with Frame Frequency.

**Note 4-7** Ensure that the LED light bar is not subjected either forward or reverse voltage while monitor set is on standby mode or not in use.

## 5 Reliability Test

AUO reliability test items are listed as following table. (Bare Panel only)

| Items                             | Condition                                                                                                     | Remark          |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------|-----------------|
| Temperature Humidity Bias (THB)   | Ta= 50°C, 80%RH, 300hours                                                                                     |                 |
| High Temperature Operation (HTO)  | Ta= 50°C, 50%RH, 300hours                                                                                     |                 |
| Low Temperature Operation (LTO)   | Ta= 0°C, 300hours                                                                                             |                 |
| High Temperature Storage (HTS)    | Ta= 60°C, 300hours                                                                                            |                 |
| Low Temperature Storage (LTS)     | Ta= -20°C, 300hours                                                                                           |                 |
| Vibration Test<br>(Non-operation) | Acceleration: 1.5 Grms<br>Wave: Random<br>Frequency: 10 - 200 Hz<br>Sweep: 30 Minutes each Axis (X, Y, Z)     |                 |
| Shock Test<br>(Non-operation)     | Acceleration: 50 G<br>Wave: Half-sine<br>Active Time: 20 ms<br>Direction: ±X, ±Y, ±Z (one time for each Axis) |                 |
| Thermal Shock Test (TST)          | -20°C/30min, 60°C/30min, 100 cycles                                                                           | <b>Note 5-1</b> |
| On/Off Test                       | On/10sec, Off/10sec, 30,000 cycles                                                                            |                 |
| ESD (Electro Static Discharge)    | Contact Discharge: ± 15KV, 150pF(330Ω ) 1sec,<br>8 points, 25 times/ point.                                   | <b>Note 5-2</b> |
|                                   | Air Discharge: ± 15KV, 150pF(330Ω ) 1sec<br>8 points, 25 times/ point.                                        |                 |
| Altitude Test                     | Operation:18,000 ft<br>Non-Operation:40,000 ft                                                                |                 |

**Note 5-1:** a. A cycle of rapid temperature change consists of varying the temperature from -20°C to 60°C and back again. Power is not applied during the test.

b. After finish temperature cycling, the unit is placed in normal room ambient for at least 4 hours before power on.

**Note 5-2:** EN61000-4-2, ESD class B: Certain performance degradation allowed

No data lost

Self-recoverable

No hardware failures.

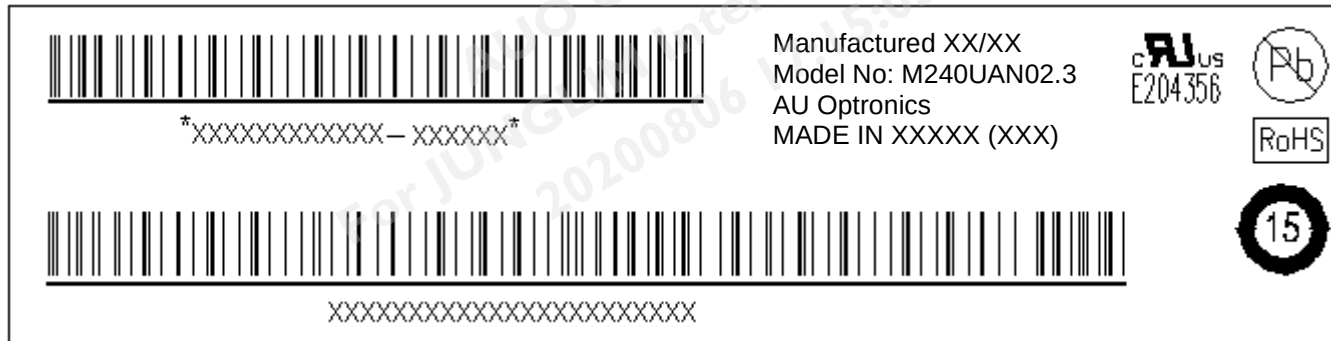
**Note 5-3:** Result Evaluation Criteria:

TFT-LCD panels test should take place after gradually cooling enough at room temperature.

In the normal application, there should be no particular problems that may affect the display function.

## 6 Shipping Label

The label is on the panel as shown below:



**Note 6-1:** For Pb Free products, AUO will add  for identification.

**Note 6-2:** For RoHS compatible products, AUO will add  for identification.

**Note 6-3:** For China RoHS compatible products, AUO will add  for identification.

**Note 6-4:** The Green Mark will be presented only when the green documents have been ready by AUO Internal Green Team.

## 7 Mechanical Characteristics

The drawing shows a rectangular display panel with a coordinate grid from 1 to 8 horizontally and A to J vertically. Key dimensions include:

- Top edge: 528.2±0.5
- Right edge: 518.4 (Active Area)
- Left edge: 342.5±0.5 (Including conductive tape)
- Inner left edge: 334 (Active Area)
- Inner top edge: 518.4 (Active Area)
- Inner bottom edge: 334 (Active Area)
- Bottom edge: 528.2±0.5
- Bottom-left corner: 13.75±0.5
- Bottom-left corner label: Conductive tape

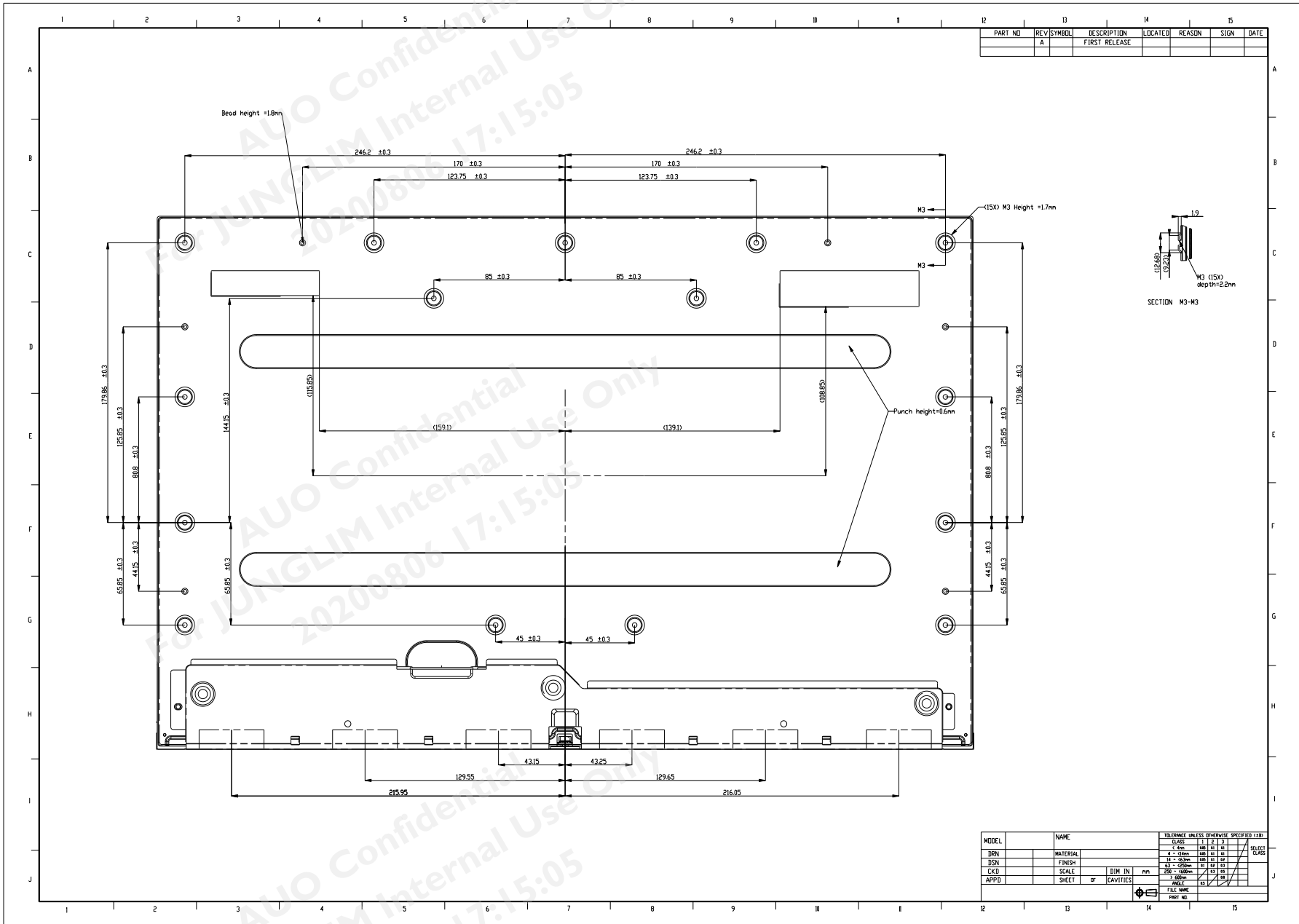
**(1) Avoid touching COF position when doing me**  
**(2) The user hole depth of the light bar side is di**

1. The dimension excludes deformation.  
2. Tolerance unspecified to be  $\pm 0.1/0.05$ .  
3. Torque of ME user hole should be within 6 kgf-cm Max. Do not rescrew more than 10 times.  
4. Avoid touching COF IC positions when doing mechanical design.  
5. Attached external parts on cell cannot quarantine the quality.  
6. Tilt & Partial disposition tolerance of Display area.  
(1) Y DIRECTION:  $-0.45^\circ/0.45^\circ$   $-0.45^\circ/0.45^\circ$   
(2) X DIRECTION:  $-0.45^\circ/0.45^\circ$   $-0.45^\circ/0.45^\circ$   
7. WHEN THIS REVERSE MODEL IS USED AS A FORWARD-TYPE MODEL (PCB ON TOP SIDE), MAY CAUSING DEFECTS TO LCM.

CELL OUTLINE  
ACTIVE AREA  
PANEL OUTLINE

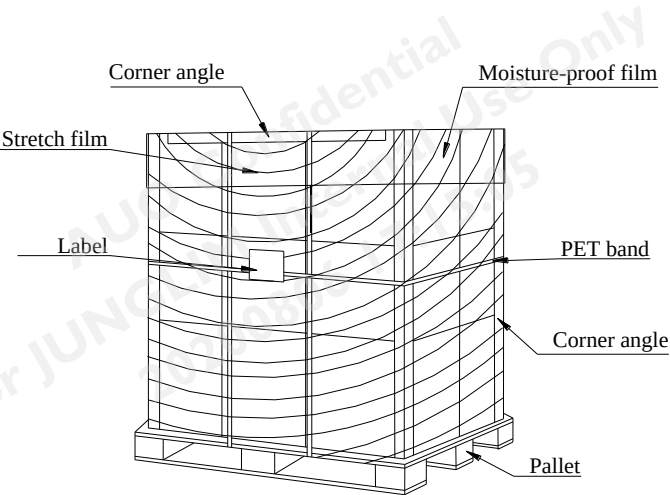
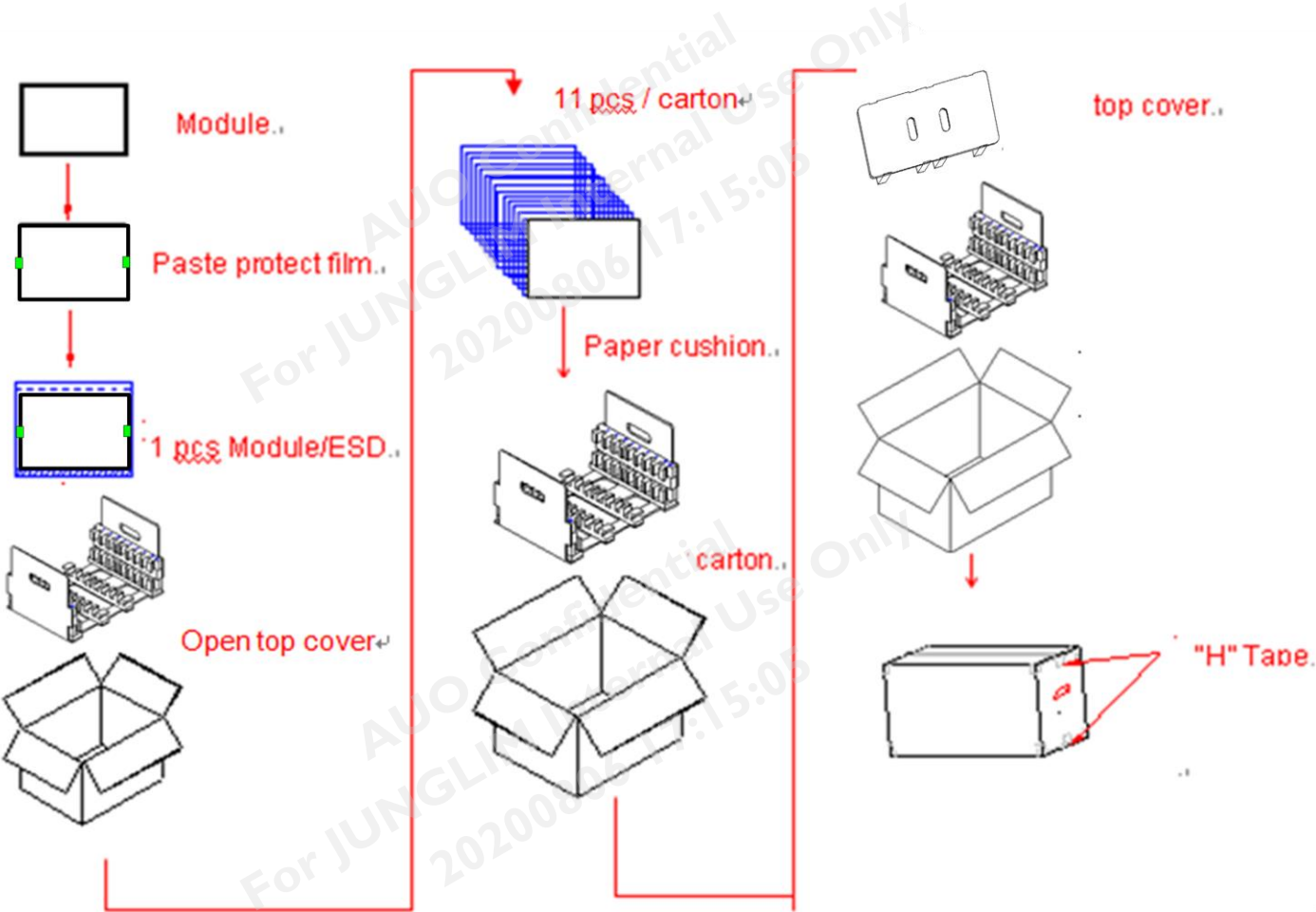






8 Packing Specification

8.1 Packing Flow



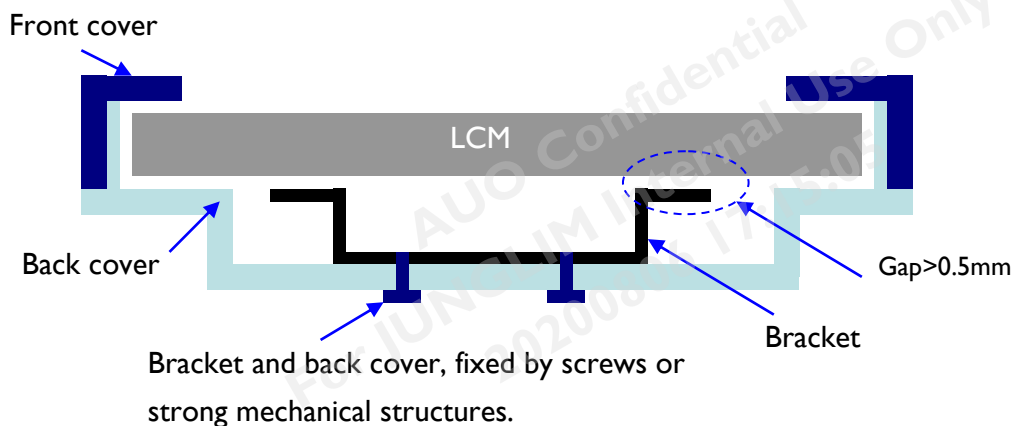
## 8.2 Pallet and shipment information

| Item                 | Specification   |                                    |            | Remark                  |
|----------------------|-----------------|------------------------------------|------------|-------------------------|
|                      | Q'ty            | Dimension                          | Weight(kg) |                         |
| Panel                | 1               | 528.2(H) × 342.4(V) × 12.5(D) Typ. | 2.4        |                         |
| Cushion              | 1               | -                                  | 0.72       |                         |
| Box                  | 1               | 617(L)mm × 362(W)mm × 420(H)mm     | 1.39       | without Panel & cushion |
| Packing Box          | 11 pcs/Box      | 617(L)mm × 362(W)mm × 420(H)mm     | 28.51      | with panel & cushion    |
| Pallet               | 1               | 1315(L)mm × 1150(W)mm × 138(H)mm   | 13.75      |                         |
| Pallet after Packing | 18 boxes/pallet | 1315(L)mm × 1150(W)mm × 1398(H)mm  | 526.93     |                         |

## 9 Design Guide for System

9.1 The gap between LCM and system rear bracket should be bigger than 0.5mm.

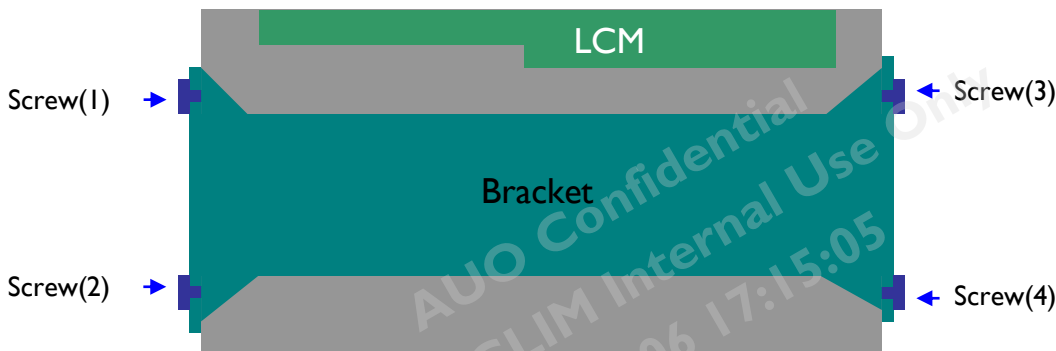
9.2 The system bracket should be fixed on back cover firmly.



9.3 The EMI gasket should be uniform and not push panel strongly.



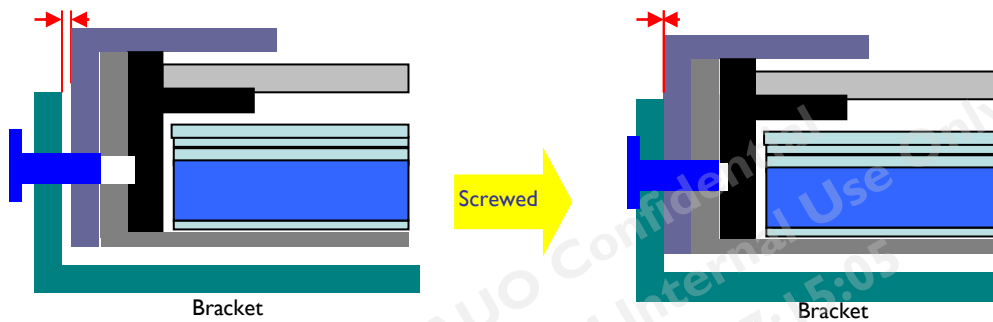
9.4 For stable assembly, the system bracket should use 4 screws to fix system and panel by dual sides.



9.5 The system bracket and panel should be in parallel with having no gap after inserting screws.

Proper and Parallel gap

0 gap and no mechanical damage



**9.6 Avoid scratching LCM, the rib on system front-cover should not exceed the bottom edge of LCM's front-bezel.**

